

Contents

1. Introduction	1
2. Theoretic Foundations	2
2.1. Probabilistic Forecasting	2
2.1.1. Prediction Space	2
2.1.2. Calibration and Sharpness	3
2.1.3. Scoring Rules and CRPS	6
2.2. Hidden Markov Models	9
2.2.1. HMM Foundations	9
2.2.2. Algorithms	11
2.2.3. HMM Forecasting	20
3. Experiment Design	23
3.1. Methodology	23
3.1.1. Preprocessing	23
3.1.2. Training	24
3.1.3. Forecasting and Evaluation	25
3.2. Data	26
3.3. Experiments	27
4. Experiment Results	29
4.1. Hyperparameter Analysis	29
4.2. Sensitivity Analysis	31
4.3. Model Analysis	32
Bibliography	35
A. Results Hyperparameter Analysis	37
B. Results Sensitivity Analysis	40
C. Results Model Analysis	42
C.1. PIT Histograms	42
C.2. Tolerance 0.1	44
C.3. Tolerance 0.01	46

1. Introduction

blabla

2. Theoretic Foundations

2.1. Probabilistic Forecasting

Generally speaking, forecasts aim to predict a future event as accurately as possible. In classic deterministic forecasting one possible outcome is predicted and the discrepancy between forecast and realization - the prediction error - is measured and in machine learning context also minimized. This approach suggests that the exact result can be predicted with a perfect prediction. However, information about future events is by nature incomplete or not fully available, and thus the events themselves are necessarily subject to uncertainty. So, it can be beneficial to acknowledge the stochastic nature of forecasts and look at the problem from a probabilistic perspective. The goal of probabilistic forecasting is to model the uncertain behavior of the future event. This can be done by approximating its probability distribution or, in a more simple form, predicting confidence intervals or quantiles of the desired distribution. A popular example are weather forecast with rain probabilities (e.g. 80% chance of rain tomorrow). In general, meteorology has been and still is the driving field of applied probabilistic forecasting, see Murphy and Winkler [17], since weather forecasts always considered meteorological events part of probabilistic processes. Nowadays, probabilistic forecasting also finds applications like e.g. in financial and economic risk management [11], prediction of low-probability events such as earthquakes [14] or floods [6].

In this section, we summarize the elementary theory of probabilistic forecasting based on the work of Gneiting and Katzfuss [10], who provides an overarching introduction to probabilistic forecasting. The section is structured into the theory of *prediction spaces*, the theoretic foundation, followed by the two most important concepts, *calibration and sharpness*, and finally concluded with *scoring* rules, the evaluation metrics for probabilistic forecasts.

2.1.1. Prediction Space

The theoretic framework for probabilistic forecasting is build on the idea of the *prediction space*, which was first introduced by Gneiting and Katzfuss [10] and developed further in Strähl and Ziegel [21]. The general problem is to predict the distribution for a random event. For our purpose sufficient, we discuss a simplified version of the prediction space based on two components.

First, we consider a one dimensional, real-valued random variable Y on the probability space $(\Omega, \mathcal{A}, P)$ with a cumulative distribution function (CDF). Y is the future event, we aim to predict, and the probability space denotes the underlying probabilistic process. The second component is the prediction described by the random variable F on the same

probability space. It takes a value in the space $\mathcal{F}$ of all continuous CDFs and measurable to the sub σ -algebra $A' \subseteq A$, hence $\mathcal{F} = \{\hat{F} \text{ is CDF} : \hat{F} \text{ continuous} \wedge \hat{F} \text{ } A'\text{-measurable}\}$. Thus, F is a random CDF depending on the information set A' , which can be interpreted as the information available to the predictor like historic data or expertise. We will use the naming 'predictive distribution' for the CDF-valued random quantity F further on. The event Y and prediction F are connected in the sense that the predictor constructs F in such a way with the available information A' that it represents the unknown distribution of Y as closely as possible. Formalizing this idea, we introduce the phrase *relative ideal* in the following definition.

Definition 2.1. The predictive distribution F is ideal relative to the information set A' if $F = \mathcal{L}(Y|A')$, where $\mathcal{L}(\cdot)$ denotes the conditional distribution of the random variable.

2.1.2. Calibration and Sharpness

In this section, we discuss beneficial properties of our predictive distribution F . Since in practice, it is not possible to observe the conditional distribution $\mathcal{L}(Y|A')$, we require different forms of assessment for the predictive distribution.

The first main idea is that the future event should occur with the same probabilities as predicted. If we could repeat the same experiment multiple times with a fixed predicted distribution $\hat{F} \in \mathcal{F}$, we would use the empiric distribution of Y and compare it to $\hat{F}$. However in general, a repetition is not possible since for each time instance the available information and thus also F changes. To overcome this problem, the *probability integral transform* (PIT) is introduced¹.

Definition 2.2. The random variable $F(Y)$ is called the PIT of the predictive distribution F .

The PIT is a transform of the original random variable in the interval $[0, 1]$ and is the value that the predictive CDF attains at the observed event. Assuming that F is the true distribution of Y , we see that the PIT $F(Y)$ follows a uniform distribution. Furthermore, if we repeat the experiment for a different time instance, the resulting PIT should still be uniformly distributed. This property is generalized by the concept of *calibration* and elaborated farther in the following definition and theorem.

Definition 2.3. Given the random variable Y and the predictive distribution F .

- (i) The predictive distribution F is called *marginally calibrated* if $\mathbb{E}[F(y)] = \mathcal{P}(Y \leq y)$.
- (ii) The predictive distribution F is called *probabilistically calibrated* if its PIT is uniformly distributed, thus $F(Y) \sim U(0, 1)$.

[Difference Marginally Calibrated vs probabilistically calibrated]

¹For the case of discrete random variables, the PIT is formulated similar to definition 2.2 with technical additions for handling discontinuities.

Theorem 2.4. If the predictive distribution F is ideal relative to the information set A' , then it is also marginally and probabilistically calibrated.

Proof. Since F is relative ideal to A' , we can describe it as $F(y) = \mathcal{P}(Y \leq y | A')$. It follows the proof for marginal calibration,

$$\mathbb{E}[F(y)] = \mathbb{E}[\mathcal{P}(Y \leq y | A')] = \mathbb{E}[\mathbb{E}[\mathbb{1}_{\{Y \leq y\}} | A']] = \mathbb{E}[\mathbb{1}_{\{Y \leq y\}}] = \mathcal{P}(Y \leq y).$$

Hereby, the characterization of the probability function via the expectancy of the indicator function $\mathbb{1}$, and the tower rule was applied.

For the probabilistic calibration, we need the fact that F takes a value in $\mathcal{F}$, which implies a continuous, monotonically increasing, surjective function. Thus, there also exists a monotonic increasing, right inverse function F^{-1} , hence

$$F \circ F^{-1} = id, \tag{2.1}$$

whereby id denotes the identity function. Considering the push-forward measure of P by Y onto $\mathbb{R}$, F is also almost surely strictly monotonic and injective. Hence, the right inverse F^{-1} is also almost surely left inverse,

$$F^{-1} \circ F = id \quad a.s. \tag{2.2}$$

It follows for $s \in [0, 1]$,

$$\begin{aligned} \mathcal{P}(F(Y) \leq s) &= \mathcal{P}(F^{-1}(F(Y)) \leq F^{-1}(s)) \\ &= \mathcal{P}(Y \leq F^{-1}(s)) \\ &= \mathbb{E}[\mathcal{P}(Y \leq F^{-1}(s) | A')] \\ &= \mathbb{E}[F(F^{-1}(s))] \\ &= \mathbb{E}[s] = s. \end{aligned}$$

Respectively, the equations are argued by the monotony of F^{-1} , equation (2.2), the tower rule, the marginal calibration of F and equation (2.1). Thus, we have shown that $F(Y)$ follows a uniform distribution. $\square$

The theorem 2.4 also holds for non-continuous random variables, which is shown in Brockwell [5]. This result provides us with a necessary, but not sufficient, condition to assess whether a prediction is possibly relatively ideal. That is, by checking the PIT for uniform distribution. There are several statistical tests to check uniformity, such as the Kolmogorov-Smirnov test or the Chi-squared test, more practical, however, is assess the PIT graphically. The standard method the histogram of the PIT of a sufficient many PIT samples². In example, each 10% interval (0-10%, 10%-20%,...) should occur in 10% of all cases, which implies that each bar of the PIT histogram should have roughly the same height at 10%. The advantage over statistical numeric test is that this approach provides additional information of the behavior of the prediction. Characteristic errors

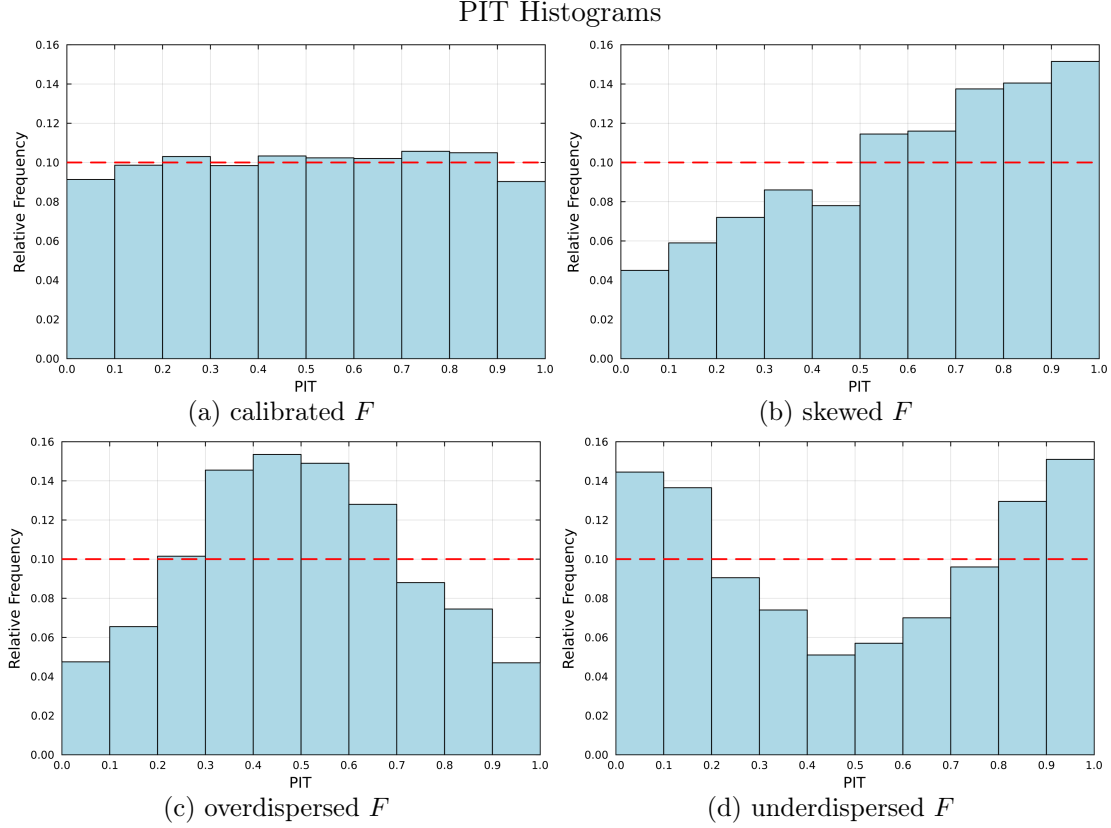


Figure 1.: PIT histogram 1a shows a uniform distributed PIT. In figure 1b the higher intervals occur more often than predicted, while lower intervals occur less often, indicating that the predictive distribution is shifted to the right compared to the underlying distribution. The PIT of figure 1c does not obtain sufficiently often the extreme ends of the distribution implying the predictive distribution to be overdispersed. Whereas in figure 1d, the opposite is the case.

such as skewness and over-/underdispersion can be easily spotted in the PIT histogram, see exemplary figures 1.

A uniformly distributed PIT, however, does not imply that the prediction F is ideal to the information set A' . Uniformity also applies for predictions which are still calibrated but do not use all of the available information. In mathematical terms, it is the case for a prediction relative ideal to another sub-sigma A^* algebras that is coarser than A' , thus $A^* \subset A'$. In example, the unconditional distribution of Y fulfills the condition of a uniformly distributed PIT and is also ideal to the trivial sub-sigma Algebra $A^* = \{\emptyset, \Omega\}$. However, it does not utilize any additional information given by A' . The question is, how

²Also, the auto-covariance function of PIT is regularly used to study historic dependencies of forecasts when using time series data.

can we assess that the maximum of available information was used?

The essential effect of utilizing more information is that uncertainty decreases, thus the predictive distribution becomes generally more concentrated. In probabilistic forecasting, the so-called attribute *sharpness* is used to describe the level of uncertainty. The "sharper" a distribution is, the less uncertain the predictive event is. Given now a set of calibrated predictive distribution, the prediction, which uses the most information, is also the one which maximizes the sharpness. Sharpness can be measured by known metrics like the variance, confidence intervals, or the entropy of a distribution, for example the normal distribution $N(0, 1)$ is sharper than $N(0, 2)$. Notably, sharpness only takes the predictive distribution into account and does not consider the true realization of the future event. Thus, it evaluates exclusively the predictive distribution.

In summarizing this section, we present the dominating principle of probabilistic forecasting formulated by Gneiting and Katzfuss [10]:

“Probabilistic forecasting has the general goal of maximizing the sharpness of the predictive distributions, subject to calibration.”

2.1.3. Scoring Rules and CRPS

This section discusses how probabilistic forecasts are evaluated numerically. It introduces the concept of *proper scoring rules* and gives a prominent example, the *continuous ranked probability score* (CRPS).

The evaluation metrics of probabilistic forecasts are not as straightforward as those applied in point forecasting. While in point forecasting, the distance between the point prediction and observed value is measured, the aim of the distribution forecast is to match the true underlying distribution, which, however, only "materializes" one observation per instance. Hence, the task is to measure the distance between distribution forecast and observation, and with sufficiently many evaluations, derive comparable results. These measures are called scoring rules, see [18] for a detailed approach.

Definition 2.5. The function $S : \mathcal{F} \times \mathbb{R} \rightarrow \mathbb{R}$ is called a scoring rule.

The scoring rule $S(\hat{F}, y)$ assigns the distribution $\hat{F}$ and the realization y of Y a numerical value. Note that in contrast to the previous section, we consider now a fixed distribution $\hat{F}$, which results from the predictive distribution F .

For the following property definition, another notation of the scoring rule is needed. Given a random variable Y , which follows the distribution $G \in \mathcal{F}$, the expression $S(\hat{F}, G)$ denotes

$$S(\hat{F}, G) = \mathbb{E}_G[S(\hat{F}, Y)].$$

In order to be applicable for the evaluation of probabilistic forecasts, a scoring rule must be *proper*.

Definition 2.6. A scoring rule is *proper* if for all $\hat{F}, G \in \mathcal{F}$

$$S(G, G) \leq S(\hat{F}, G)$$

holds. It is *strictly proper* if $S(G, G) = S(\hat{F}, G)$ implies $\hat{F} = G$.

Thus, a proper scoring rule is designed in such a way that for the true underlying distribution the score is minimized in expectation.

In probabilistic load forecasting, the Continuous Ranked Probability Score (CRPS) has asserted itself in many applications, notably here is the Global Energy Forecasting Competition (GEFCom2014) summarized in Hong, Pinson, Fan, Zareipour, Troccoli, and Hyndman [13]. Its original form is defined by

$$\mathbf{CRPS}(\hat{F}, y) = \int_{-\infty}^{\infty} (\hat{F}(x) - \mathbb{1}_{\{y < x\}})^2 dx.$$

The elementary idea is to calculate on the left side of the observation the area underneath the predicted distribution and on the right side the area above the predicted distribution, see figure 2. The value is minimized, if the observation lies "central" and the distribution is as sharp as possible. In this way, the CRPS takes the two major concepts of probabilistic forecasting, calibration and sharpness, into account.

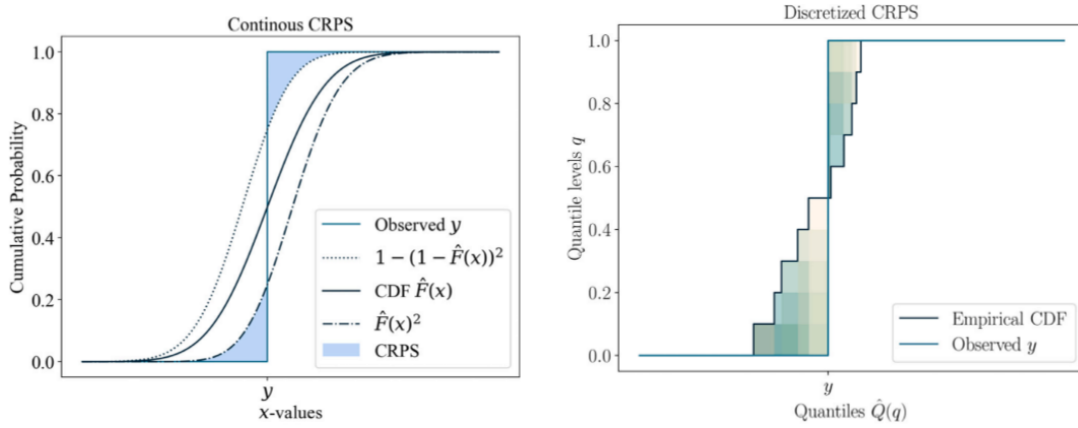


Figure 2.: Graphical example of the continuous and discretized CRPS, reproduced from Botman, Lago, Becker, Vanthournout, and Moor [3]

However, computing the continuous version of the CRPS can be computationally expensive. For this reason, we use a discretized version of the CRPS which is also applied for quantile predictions. This is formulated together with another popular scoring rule, the Pinball score,

$$\mathbf{CRPS}(\hat{F}, y) = \sum_q \mathbf{Pinball}(\hat{Q}(q), y, q)$$

with $\mathbf{Pinball}(\hat{Q}(q), y, q)$:

$$= \begin{cases} (1 - q) \cdot (\hat{Q}(q) - y), & \text{if } y < \hat{Q}(q) \\ q \cdot (y - \hat{Q}(q)), & \text{if } y \geq \hat{Q}(q) \end{cases}$$

where q denotes a quantile and $\hat{Q}(q)$ the predicted value for q of the distribution prediction. For each evaluation, we calculate the quantiles 0, 0.01, 0.02, ... 0.99, 1 from our predicted distribution and compute the CRPS. To achieve comparable results, multiple evaluations are needed and then the mean CRPS is taken.

2.2. Hidden Markov Models

In simple words, a Hidden Markov Model (HMM) is a stochastic process which underlies a hidden Markov process but can only be observed by probabilistic signals emitted by states of this hidden process. Originally, it was first studied by Baum and Petrie [2] as *probabilistic functions of Markov chains* and gained significance during the 1980s and 1990s as application in speech recognition, see Rabiner [19]. During this time, HMMs dominated automated speech recognition research and commercial application, forming the backbone of early voice-controlled interfaces. In general, HMMs are applied for tasks that require decoding, recognition and classification like DNA analysis and gene recognition [15, 22] or behavior detection [4]. Recently, Makonin, Popowich, Bajić, Gill, and Bartram [16] used HMM to disaggregate electric power load profiles and identify the usage of single electric devices. In the last years however, machine learning techniques like recurrent neural networks or transformers have superseded HMMs especially in large-scale tasks of decoding and classification. Nevertheless, HMM still remains a valuable tool in many applications due to its interpretability, mathematical tractability and efficiency in low-data settings.

In this thesis, our aim is to investigate another possible application of HMM, namely forecasting. In the existing literature, there have been only a few single studies conducted using HMM in forecasting [8, 9, 23]. However, to the best of the author’s knowledge, these studies have only applied HMM for deterministic predictions and for small-scale models (less than 10 hidden states). Additionally, the papers lack an exhaustive performance analysis of different parameter settings. Applying HMMs for probabilistic forecasting offers a great opportunity because one of the biggest challenges in probabilistic forecasting is its computational effort, while HMMs probabilistic forecasts are available very efficiently due to its stochastic design. Additionally, with today’s computer, models up to a couple of hundred hidden states are possible to calculate.

For this reason, we present in this section a theoretic approach to generate probabilistic forecasts using HMM and discuss the necessary theory and existing algorithms.

2.2.1. HMM Foundations

Starting with the notion of HMM, there are N possible hidden states forming the *state space* S and M different observation values forming the *observation space* O ,

$$\begin{aligned} S &= \{s_1, s_2, \dots, s_N\} \\ O &= \{o_1, o_2, \dots, o_M\}. \end{aligned}$$

We restrict the discussion to discrete HMM with finite state and observation space, thus $N, M \in \mathbb{N}$. Denote the sets of indexes by $I_S = \{1, \dots, N\}$ and $I_O = \{1, \dots, M\}$. Let $T \in \mathbb{N}$ be the time horizon and its corresponding index set $I_T = \{1, \dots, T\}$. The HMM consists of two dependent stochastic processes, a hidden and an observable one. For each time instance, the hidden process $\{Z_t, t \in I_T\}$ takes on values in the state space, while the

observable process $\{X_T, t \in I_T\}$ is located in the observation space,

$$\begin{aligned}(Z_1, Z_2, \dots, Z_T) &\in S^T \\ (X_1, X_2, \dots, X_T) &\in O^T.\end{aligned}$$

To distinguish between random variables and the event of random variables taking fixed values, we introduce the following compact notation. The event of Z_t taking a value is denoted by $\dot{Z}_t$,

$$\dot{Z}_t = \{Z_t = s_{i_t}\},$$

for a fixed $i_t \in I_S$ that describes the state index at time t . All events until time instance t summed up by the event vector $\dot{\mathbf{Z}}_t$,

$$\dot{\mathbf{Z}}_t = (\dot{Z}_1, \dot{Z}_2, \dots, \dot{Z}_t)$$

Analogously for the observable process, the fixed $i_t \in I_S$ describes the observation index at time t ,

$$\dot{X}_t = \{X_t = o_{k_t}\},$$

In addition to $\dot{\mathbf{X}}_t$, the vector $\dot{\mathbf{X}}'_t$ describes all occurred events after time instance t (including time instance t),

$$\begin{aligned}\dot{\mathbf{X}}_t &= (\dot{X}_1, \dot{X}_2, \dots, \dot{X}_t) \\ \dot{\mathbf{X}}'_t &= (\dot{X}_t, \dot{X}_{t+1}, \dots, \dot{X}_{T-1}, \dot{X}_T).\end{aligned}$$

The HMM is entirely characterized through the following three properties.

(A1) Markov property:

$$\mathcal{P}(\dot{Z}_{t+1} | \dot{\mathbf{Z}}_t) = \mathcal{P}(\dot{Z}_{t+1} | \dot{Z}_t)$$

(A2) Homogeneity:

$$\mathcal{P}(Z_{t+1} = s_j | Z_t = s_i) = \mathcal{P}(Z_t = s_j | Z_{t-1} = s_i) = \dots = \mathcal{P}(Z_2 = s_j | Z_1 = s_i)$$

(A3) Independent observation assumption:

$$\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{t-1}, \dot{\mathbf{X}}'_{t+1}, \dot{\mathbf{Z}}_T) = \mathcal{P}(\dot{X}_t | \dot{Z}_t)$$

The properties **A1** and **A2** determine the process Z to be a homogeneous Markov chain. Thus, a future event depends on the past events only through the present event. The third property **A3** implies that the observation is influenced only by the current hidden state. Due to these properties, a HMM can be parameterized by stochastic matrices and vectors similar to the transition matrix of standard Markov chains. The hidden Markov

process $\{Z_t, t \in I_T\}$ is determined by the $N \times N$ -dimensional *transition matrix* $\mathbb{A}$ with elements a_{ij} ,

$$a_{ij} = \mathcal{P}(Z_t = s_j | Z_{t-1} = s_i) \quad \text{for } i, j = 1, \dots, N,$$

and by the N -dimensional *initial distribution vector* π with

$$\pi_i = \mathcal{P}(Z_1 = s_i) \quad \text{for } i = 1, \dots, N.$$

The $N \times M$ -dimensional *observation matrix* $\mathbb{B}$ describes the emission probabilities of the observation process $\{X_t, t \in I_T\}$ with elements b_{ik} ,

$$b_{ik} = \mathcal{P}(X_t = o_k | Z_t = s_i) \quad \text{for } i = 1, \dots, N \text{ and } k = 1, \dots, M.$$

These three parameters fully describe a finite HMM, summarized by $\lambda = (\mathbb{A}, \mathbb{B}, \pi)$.

2.2.2. Algorithms

The three strong properties [A1](#), [A2](#), [A3](#) allow us to compute HMM related problems very efficiently. Before we present common algorithms, we first identify the three main questions regarding HMM formulated by Rabiner [\[19\]](#):

1. **Evaluation:**

Given an HMM and observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, what is the probability that this sequence occurred,

$$\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})?$$

2. **Decoding:**

Given a HMM and observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, what is the most probable state sequence $\dot{\mathbf{Z}}_{\mathbf{T}}$ which emits the given observations,

$$\arg \max_{\dot{\mathbf{Z}}_{\mathbf{T}}} \mathcal{P}(\dot{\mathbf{Z}}_{\mathbf{T}} | \dot{\mathbf{X}}_{\mathbf{T}})?$$

3. **Training:**

Given observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, what are the optimal parameters λ of the HMM to describe the given observations,

$$\arg \max_{\lambda} \mathcal{P}_{\lambda}(\dot{\mathbf{X}}_{\mathbf{T}})?$$

Due to the three properties of HMMs, all of these quantities can be efficiently computed. In this section, we discuss solutions and algorithms for the problem of evaluation and training, which is relevant for the application of HMM as a forecasting tool. The question of decoding can also be relevant for the analysis of existing forecasting models to develop a deeper understanding of the underlying mechanisms, which however we do not discuss

further in this thesis.

To show the computational complexity of the above problems, we first make an exemplary naive approach of calculating the evaluation problem. Given $\dot{\mathbf{X}}_{\mathbf{T}}$, we reformulate the desired probability using the conditional probability over all possible hidden state sequences $\mathcal{Z}_{\mathcal{T}}$,

$$\begin{aligned} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \mathcal{Z}_{\mathcal{T}}} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}, \dot{\mathbf{Z}}_{\mathbf{T}}) \\ &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \mathcal{Z}_{\mathcal{T}}} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}} | \dot{\mathbf{Z}}_{\mathbf{T}}) \mathcal{P}(\dot{\mathbf{Z}}_{\mathbf{T}}). \\ &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \mathcal{Z}_{\mathcal{T}}} \left(\prod_{t=1}^T \mathcal{P}(\dot{X}_t | \dot{Z}_t) \right) \left(\mathcal{P}(\dot{Z}_1) \prod_{t=2}^T \mathcal{P}(\dot{Z}_t | \dot{Z}_{t-1}) \right) \\ &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \mathcal{Z}_{\mathcal{T}}} \left(\prod_{t=1}^T b_{i_t k_t} \right) \left(\pi_{i_1} \prod_{t=2}^T a_{i_{t-1} i_t} \right). \end{aligned}$$

The set of all possible hidden state sequences is defined formally by

$$\mathcal{Z}_{\mathcal{T}} = \{(Z_t = s_{i_t})_{t \in \{1, \dots, T\}} \text{ for } (i_t)_{t \in \{1, \dots, T\}} \in I_S^T\}.$$

Since I_S has N elements, there are N^T different possible hidden state sequences. Thus, evaluating $\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$ has a non-polynomial complexity of order $\mathcal{O}(N^T \cdot 2T)$ and for larger models or longer time horizon reaches quickly the limits of computational feasibility. However, due to its Markovian behavior, the evaluation of $\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$ can be done more efficiently by building up an iterative procedure based on its three properties. This is called the *forward algorithm* presented in the following.

We define the quantity $\alpha_t(i)$ as

$$\alpha_t(i) = \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_i), \quad \text{for } i \in I_S, t \in \{1, \dots, T\}.$$

For $t \geq 2$, the joint probability $\alpha_t(j)$ can be expressed by its predecessors $\alpha_{t-1}(\cdot)$ and the parameters of the given HMM,

$$\begin{aligned} \alpha_t(j) &= \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_j) \\ &= \sum_{i \in I_S} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_j, Z_{t-1} = s_i) \\ &= \sum_{i \in I_S} \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t-1}}, Z_t = s_j, Z_{t-1} = s_i) \mathcal{P}(Z_t = s_j | \dot{\mathbf{X}}_{\mathbf{t-1}}, Z_{t-1} = s_i) \\ &\quad \cdot \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t-1}}, Z_{t-1} = s_i) \\ &= \sum_{i \in I_S} \mathcal{P}(\dot{X}_t | Z_t = s_j) \mathcal{P}(Z_t = s_j | Z_{t-1} = s_i) \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t-1}}, Z_{t-1} = s_i) \\ &= \sum_{j \in I_S} b_{jk_t} \cdot a_{ij} \cdot \alpha_{t-1}(i). \end{aligned}$$

The computation of the first joint probabilities $a_1(\cdot)$ requires the initial distribution π ,

$$\alpha_1(i) = \mathcal{P}(\dot{X}_1, Z_1 = s_i) = \mathcal{P}(\dot{X}_1 | Z_1 = s_i) \mathcal{P}(Z_1 = s_i) = b_{ik_1} \cdot \pi_i.$$

Using the quantities $\alpha_T(\cdot)$, the probability targeted in the evaluation problem is calculated by

$$\mathcal{P}(\dot{\mathbf{X}}_T) = \sum_{i \in I_S} \mathcal{P}(\dot{\mathbf{X}}_T, Z_T = s_i) = \sum_{i \in I_S} \alpha_T(i).$$

This is the standard forward algorithm, discussed in [19], and has a computational complexity of order $\mathcal{O}(N^2 \cdot 3T)$, which is significantly lower than that of the naive approach. However, this algorithm also has its drawbacks in practice. If we look closely at the definition of the joint probability $\mathcal{P}(\dot{\mathbf{X}}_T)$, it can be observed that with increasing time horizon T this probability is becoming smaller and in regular cases converges to 0. It implies that with a significantly large time horizon, this quantity will be smaller than any for a computer representable number and thus not feasible to compute. For this reason, we present a version of the forward algorithm adapted to conditional probabilities which do not vanish with increasing T and allows for the computation of the log-likelihood of the evaluation problem.

We define two quantities $\bar{\alpha}$ and $\tilde{\alpha}$ similar to the original α ,

$$\bar{\alpha}_t(i) = \mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{t-1})$$

and

$$\tilde{\alpha}_t(i) = \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_t),$$

for $i \in I_S$ and $t \in \{1, \dots, T\}$. On the one hand, $\bar{\alpha}_t(i)$ is computed by the predecessors $\tilde{\alpha}_{t-1}(\cdot)$,

$$\begin{aligned} \bar{\alpha}_t(i) &= \mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{t-1}) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t, Z_t = s_i, Z_{t-1} = s_j | \dot{\mathbf{X}}_{t-1}) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{t-1}, Z_{t-1} = s_j) \mathcal{P}(Z_{t-1} = s_j | \dot{\mathbf{X}}_{t-1}) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t | Z_t = s_i) \mathcal{P}(Z_t = s_i | Z_{t-1} = s_j) \mathcal{P}(Z_{t-1} = s_j | \dot{\mathbf{X}}_{t-1}) \\ &= \sum_{j \in I_S} b_{ikt} \cdot a_{ji} \cdot \tilde{\alpha}_{t-1}(j). \end{aligned}$$

Considering the fact of

$$\begin{aligned} \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{t-t}) &= \sum_{i \in I_S} \mathcal{P}(Z_t = s_i, \dot{X}_t | \dot{\mathbf{X}}_{t-t}) \\ &= \sum_{i \in I_S} \bar{\alpha}_i(t), \end{aligned}$$

we can express on the other hand $\tilde{\alpha}_t(i)$ by $\bar{\alpha}_t(\cdot)$,

$$\begin{aligned}\tilde{\alpha}_t(i) &= \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}}) \\ &= \frac{\mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-\mathbf{t}})}{\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-\mathbf{t}})} \\ &= \frac{\bar{\alpha}_t(i)}{\sum_{j \in I_S} \bar{\alpha}_j(t)}.\end{aligned}$$

As mentioned, the probability $\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$ induces underflow errors. However, the log-likelihood of the evaluation problem, $\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$ lies generally within the computationally feasible range. The log-likelihood is calculated by splitting up the original problem into subproblems,

$$\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) = \log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) = \log \prod_{t=1}^T \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}) = \sum_{t=1}^T \log \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}),$$

and using the fact that $\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-\mathbf{t}}) = \sum_{j \in I_S} \bar{\alpha}_j(t)$ as shown above. Additionally, consider the following relation between $\tilde{\alpha}$ and α ,

$$\begin{aligned}\tilde{\alpha}_t(i) &= \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}}) \\ &= \frac{\mathcal{P}(Z_t = s_i, \dot{\mathbf{X}}_{\mathbf{t}})}{\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}})} \\ &= \frac{\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_i)}{\sum_{j \in I_S} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_j)} \\ &= \frac{\alpha_t(i)}{\sum_{j \in I_S} \alpha_j(t)}.\end{aligned}$$

This shows that $\tilde{\alpha}_t(i) = c_t \cdot \alpha_t(i)$ with the constant factor $c_t = \frac{1}{\sum_{j \in I_S} \alpha_t(j)}$. Thus, $\tilde{\alpha}_t$ is a scaled version of α_t , which becomes relevant in the training problem.

Adapted forward algorithm

1. *Initialization:* ($t = 1$)

$$\bar{\alpha}_1(i) = b_{ik_1} \cdot \pi_i \quad \text{for } i \in I_S$$

$$\mathcal{P}(\dot{X}_1) = \sum_{j \in I_S} \bar{\alpha}_1(j)$$

$$\tilde{\alpha}_1(i) = \bar{\alpha}_1(i) / \mathcal{P}(\dot{X}_1) \quad \text{for } i \in I_S$$
 2. *Iteration:* ($t = 2, \dots, T$)

$$\bar{\alpha}_t(i) = \sum_{j \in I_S} b_{ik_t} \cdot a_{ji} \cdot \tilde{\alpha}_{t-1}(j) \quad \text{for } i \in I_S$$

$$\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{t-1}) = \sum_{j \in I_S} \bar{\alpha}_t(j)$$

$$\tilde{\alpha}_t(i) = \bar{\alpha}_t(i) / \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{t-1}) \quad \text{for } i \in I_S$$
 3. *Termination:*

$$\log \mathcal{P}(\dot{\mathbf{X}}_T) = \sum_{t=1}^T \log \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{t-1})$$
- return* $\log P(\dot{\mathbf{X}}_T)$

In addition, the evaluation can also be computed with the so-called *backward algorithm*. It utilizes the HMM properties similar as the forward algorithm but iterates from the other temporal direction, thus backwards. Both of these algorithms are needed later on to solve the training problem.

Let $\beta_t(i)$ be the following quantity for $t \in \{1, \dots, T-1\}$ and $i \in I_S$,

$$\beta_t(i) = \mathcal{P}(\dot{\mathbf{X}}'_{t+1} | Z_t = s_i).$$

The probabilities $\beta_t(\cdot)$ are described recursively and initiated by setting $\beta_T(\cdot)$ to one,

$$\beta_T(i) = 1.$$

Then,

$$\begin{aligned} \beta_t(i) &= \mathcal{P}(\dot{X}_{t+1}, \dot{\mathbf{X}}'_{t+2} | Z_t = s_i) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_{t+1}, \dot{\mathbf{X}}'_{t+2} | Z_t = s_i, Z_{t+1} = s_j) \mathcal{P}(Z_{t+1} = s_j | Z_t = s_i) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_{t+1} | Z_{t+1} = s_j) \mathcal{P}(\dot{\mathbf{X}}'_{t+2} | Z_{t+1} = s_j) \mathcal{P}(Z_{t+1} = s_j | Z_t = s_i) \\ &= \sum_{j \in I_S} b_{jk_{t+1}} \cdot \beta_{t+1}(j) \cdot a_{ij}. \end{aligned}$$

Since $\dot{\mathbf{X}}_T = \dot{\mathbf{X}}'_1$, the desired probability of the evaluation problem is then computed by,

$$\begin{aligned}
 \mathcal{P}(\dot{\mathbf{X}}_T) &= \mathcal{P}(\dot{\mathbf{X}}'_1) \\
 &= \sum_{i \in I_S} \mathcal{P}(\dot{\mathbf{X}}'_1 | Z_1 = s_i) \mathcal{P}(Z_1 = s_i) \\
 &= \sum_{i \in I_S} \mathcal{P}(\dot{X}_1 | Z_1 = s_i) \mathcal{P}(\dot{\mathbf{X}}'_2 | Z_1 = s_i) \mathcal{P}(Z_1 = s_i) \\
 &= \sum_{i \in I_S} b_{ik_1} \cdot \beta_1(j) \cdot \pi_i.
 \end{aligned}$$

Similar to the forward algorithm, the standard version of the backward algorithm as described above induces numerical difficulties in practice and therefore, demands an adaptation. To avoid infeasibility, $\beta_t(\cdot)$ is scaled with the constant $c_t = \sum_{j \in I_S} \beta_t(j)$ for $t \in \{1, \dots, T-1\}$ and $c_T = 1$. summarized by $\tilde{\beta}_t(\cdot)$,

$$\tilde{\beta}_t(i) = \frac{\beta_t(i)}{c_t} = \frac{\mathcal{P}(\dot{\mathbf{X}}'_{t+1} | Z_t = s_i)}{\sum_{j \in I_S} \mathcal{P}(\dot{\mathbf{X}}'_{t+1} | Z_t = s_j)}$$

This scaling implies that

$$\sum_{i \in I_S} \tilde{\beta}_t(i) = 1,$$

and thus assures feasibility for all $\tilde{\beta}_t(i)$. The factor c_t has no meaningful interpretation as the scaling factor of the adapted forward algorithm and also cannot be directly computed. However, due to the above result, the ratio c_t/c_{t+1} is calculated as follows,

$$\begin{aligned}
 1 &= \sum_{i \in I_S} \tilde{\beta}_t(i) \\
 &= \sum_{i \in I_S} \frac{\beta_t(i)}{c_t} \\
 &= \sum_{i \in I_S} \frac{\sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \beta_{t+1}(j)}{c_t} \\
 &= \sum_{i \in I_S} \frac{\sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j) \cdot c_{t+1}}{c_t} \\
 &= \frac{c_{t+1}}{c_t} \sum_{i \in I_S} \sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j).
 \end{aligned}$$

Hence,

$$\frac{c_t}{c_{t+1}} = \sum_{i \in I_S} \sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j).$$

The log-likelihood is then derived by

$$\begin{aligned}
 \log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) &= \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \beta_1(j) \right) \\
 &= \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \tilde{\beta}_1(j) \cdot c_1 \right) \\
 &= \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \tilde{\beta}_1(j) \cdot \prod_{t=1}^{T-1} \frac{c_t}{c_{t+1}} \right) \\
 &= \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \tilde{\beta}_1(j) \right) + \sum_{t=1}^{T-1} \log \frac{c_t}{c_{t+1}}.
 \end{aligned}$$

Adapted backward algorithm

1. *Initialization:* ($t = T$)
 $\tilde{\beta}_T(i) = 1$
 $c_T = 1$
 2. *Iteration:* ($t = T - 1, \dots, 1$)
 $\frac{c_t}{c_{t+1}} = \sum_{i \in I_S} \sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j)$
 $\beta_t(i) = \sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j)$
 3. *Termination:*
 $\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) = \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \tilde{\beta}_1(j) \right) + \sum_{t=1}^{T-1} \log \frac{c_t}{c_{t+1}}$
- return* $\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$

Turning now to the more advanced problem of training HMMs, we introduce the *Baum-Welch algorithm*. The objective is to find the optimal parameters of an HMM to fit a given observation sequence. The Baum-Welch algorithm is a special case of the expectation maximization (EM) algorithm, which is an iterative approach to find the local maxima of the likelihood for the parameter settings of a statistical model. Each iteration is divided into two steps, the expectation and the maximization step. While in the expectation step, the current expectational values under fixed parameters are computed, these expectational values are then used in the maximization step to re-estimate the next parameters. The reader is referred to Dempster, Laird, and Rubin [7] for a detailed discussion of EM algorithms and to Baum [1] for a formal derivation of the Baum-Welch algorithm. In this work, we illustrate the main ideas of Baum-Welch and give an intuitional approach. Let $\xi_t(i, j)$ be the conditional probability of being in the hidden state s_i at the time instance t and transitioning to state s_j under the given observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$,

$$\xi_t(i, j) = \mathcal{P}(Z_t = s_i, Z_{t+1} = s_j \mid \dot{\mathbf{X}}_{\mathbf{T}}).$$

Due to the fact of

$$\begin{aligned}
 \mathcal{P}(Z_t = s_i, Z_{t+1} = s_j, \dot{\mathbf{X}}_{\mathbf{T}}) &= \\
 &= \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}} | Z_t = s_i, Z_{t+1} = s_j) \mathcal{P}(Z_t = s_i, Z_{t+1} = s_j) = \\
 &= \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}} | Z_t = s_i) \mathcal{P}(\dot{\mathbf{X}}'_{t+1} | Z_{t+1} = s_j) \mathcal{P}(Z_{t+1} = s_j | Z_t = s_i) \mathcal{P}(Z_t = s_i) \\
 &= \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}, Z_t = s_i) \mathcal{P}(\dot{\mathbf{X}}'_{t+1} | Z_{t+1} = s_j) \mathcal{P}(\dot{X}_{t+1} | Z_{t+1} = s_j) \mathcal{P}(Z_{t+1} = s_j | Z_t = s_i) \\
 &= \alpha_t(i) \cdot \beta_{t+1}(j) \cdot b_{jk_{t+1}} \cdot a_{ij},
 \end{aligned}$$

we can express $\xi_t(i, j)$ as the following,

$$\begin{aligned}
 \xi_t(i, j) &= \frac{\mathcal{P}(Z_t = s_i, Z_{t+1} = s_j, \dot{\mathbf{X}}_{\mathbf{T}})}{\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})} \\
 &= \frac{\mathcal{P}(Z_t = s_i, Z_{t+1} = s_j, \dot{\mathbf{X}}_{\mathbf{T}})}{\sum_{i' \in I_S} \sum_{j' \in I_S} \mathcal{P}(Z_t = s_{i'}, Z_{t+1} = s_{j'}, \dot{\mathbf{X}}_{\mathbf{T}})} \\
 &= \frac{\alpha_t(i) \cdot a_{ij} \cdot b_{jk_{t+1}} \cdot \beta_{t+1}(j)}{\sum_{i' \in I_S} \sum_{j' \in I_S} \alpha_t(i') \cdot a_{i'j'} \cdot b_{j'k_{t+1}} \cdot \beta_{t+1}(j')}.
 \end{aligned}$$

Where the quantities $\alpha_t(i)$ and $\beta_t(j)$ are used from the forward and backward algorithm. Since $\alpha_t(\cdot)$ and $\beta_t(\cdot)$ are both used in the nominator as well as the denominator, they can be exchanged by its scaled versions $\tilde{\alpha}_t(\cdot)$ and $\tilde{\beta}_t(\cdot)$.

Further, consider the probability of being in state i during the time instance t given $\dot{\mathbf{X}}_{\mathbf{T}}$, thus³

$$\gamma_t(i) = \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{T}}) = \sum_{j \in I_S} \mathcal{P}(Z_t = s_i, Z_{t+1} = s_j | \dot{\mathbf{X}}_{\mathbf{T}}) = \sum_{j \in I_S} \xi_t(i, j).$$

Given the quantities $\xi_t(\cdot, \cdot)$ and $\gamma_t(\cdot)$, useful information on the hidden process is available. The expected number of transitions from state s_i to state s_j is equivalent to the sum of all $\xi_t(i, j)$ over $t \in \{1, \dots, T-1\}$,

$$\mathbb{E}(\#transitions \text{ from } s_i \text{ to } s_j) = \sum_{t=1}^{T-1} \xi_t(i, j).$$

The expected number of visits in state s_i is described by the sum of all $\gamma_t(i)$ over $t \in \{1, \dots, T\}$,

$$\mathbb{E}(\#visits \text{ in } s_i) = \sum_{t=1}^T \gamma_t(i).$$

Similarly, the expected number of transitions from state s_i is the sum of all $\gamma_t(i)$ over $t \in \{1, \dots, T-1\}$,

$$\mathbb{E}(\#transitions \text{ from } s_i) = \sum_{t=1}^{T-1} \gamma_t(i).$$

³In case $t = T$, $\gamma_T(j)$ is computed by $\sum_{i \in I_S} \xi_{T-1}(i, j)$.

The above equations represent the expectation step of the Baum-Welch algorithm. In the second step, the maximization step, the parameters of the HMM are re-estimated. Hereby, the results from the expectation step and the probabilistic concept of counting occurrences are used the following way,

$$\begin{aligned}\bar{\pi}_i &= \mathcal{P}(Z_1 = s_i | \dot{\mathbf{X}}_{\mathbf{T}}) = \gamma_1(i) \\ \bar{a}_{ij} &= \frac{\mathbb{E}(\# \text{transitions from } s_i \text{ to } s_j)}{\mathbb{E}(\# \text{transitions from } s_i)} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \\ \bar{b}_{ik} &= \frac{\mathbb{E}(\# \text{visits in } s_i \text{ and observe } o_k)}{\mathbb{E}(\# \text{visits in } s_i)} = \frac{\sum_{t=1}^T \mathbb{1}_{\{o_{k_t}=o_k\}} \gamma_t(i)}{\sum_{t=1}^T \gamma_t(i)}.\end{aligned}$$

Baum [1] shows that the likelihood of the re-estimated parameters $\bar{\lambda} = (\bar{\pi}, \bar{A}, \bar{B})$ is always greater or equal to the likelihood of the previous parameter setting λ . By iterating this procedure, the parameters are improved and thus, provides an applicable training method. However, it only converges against a local minimum, not guaranteeing an optimal solution to the training problem. Nonetheless, it has been shown in practice that in general this local minimum is sufficient and additionally not very sensitive to different initial settings. One possible reason is that the parameter space is highly symmetric due to interchangeability of the hidden states. Therefore, we choose the initial parameters at random. The algorithm terminates if the log-likelihood has converged or a maximum number of iterations is reached.

Baum-Welch algorithm

1. Initialization

$\lambda = (\pi, \mathbb{A}, \mathbb{B})$ is chosen randomly

For i in $1, \dots, \text{maxIteration}$:

2. Expectation step

$$\begin{aligned}\xi_t(i, j) &= \frac{\tilde{\alpha}_t(i) \cdot a_{ij} \cdot b_{j_{k_{t+1}}} \cdot \tilde{\beta}_{t+1}(j)}{\sum_{i' \in I_S} \sum_{j' \in I_S} \tilde{\alpha}_t(i') \cdot a_{i'j'} \cdot b_{j'_{k_{t+1}}} \cdot \tilde{\beta}_{t+1}(j')} \\ \gamma_t(i) &= \sum_{j \in I_S} \xi_t(i, j)\end{aligned}$$

3. Maximization step

$$\begin{aligned}\bar{\pi}_i &= \gamma_1(i) \\ \bar{a}_{ij} &= \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \\ \bar{b}_{ik} &= \frac{\sum_{t=1}^T \mathbb{1}_{\{o_{k_t}=o_k\}} \gamma_t(i)}{\sum_{t=1}^T \gamma_t(i)}\end{aligned}$$

4. Termination condition

if $\mathcal{P}_{\lambda}(\dot{\mathbf{X}}_{\mathbf{T}}) = \mathcal{P}_{\bar{\lambda}}(\dot{\mathbf{X}}_{\mathbf{T}})$: terminate
else: set $\lambda = \bar{\lambda}$

return λ

2.2.3. HMM Forecasting

This section introduces the theoretic reasoning behind HMM forecasting methods. Given the efficient algorithms of the previous section, this task is derived straight forward. Additional, while we aim to use the resulting probabilities for probabilistic forecasts, we show possible methods to apply HMM in point forecasting.

The general goal of the probabilistic forecasting problem is to predict with which probabilities future events are occurring. Specifying a fixed future time horizon $H \in \mathbb{N}$ and given the historic observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, we aim to calculate the probability

$$\mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}})$$

for all possible observations $k \in I_O$. Due to the HMM properties, these probabilities are calculated by tracing back all possible hidden state sequences probabilities leading up to the forecasting time instance.

First, we break down the initial probability into the following,

$$\mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}}) = \sum_{j \in I_S} \mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}}, Z_{T+H} = s_j) \mathcal{P}(Z_{T+H} = s_j | \dot{\mathbf{X}}_{\mathbf{T}}) \quad (2.3)$$

$$= \sum_{j \in I_S} b_{jo_k} \mathcal{P}(Z_{T+H} = s_j | \dot{\mathbf{X}}_{\mathbf{T}}). \quad (2.4)$$

Hereby, we observe that we can reduce the original problem to previous hidden state probabilities. Further, these hidden state probabilities are also reducible to its predecessors,

$$\mathcal{P}(Z_t = s_j | \dot{\mathbf{X}}_{\mathbf{T}}) = \sum_{i \in I_S} \mathcal{P}(Z_t = s_j | \dot{\mathbf{X}}_{\mathbf{T}}, Z_{t-1} = s_i) \mathcal{P}(Z_{t-1} = s_i | \dot{\mathbf{X}}_{\mathbf{T}}) \quad (2.5)$$

$$= \sum_{i \in I_S} a_{ij} \cdot \mathcal{P}(Z_{t-1} = s_i | \dot{\mathbf{X}}_{\mathbf{T}}) \quad (2.6)$$

for all $t > T$. For the case of $t = T$, the quantity $\mathcal{P}(Z_T = s_i | \dot{\mathbf{X}}_{\mathbf{T}})$ is already known from the adapted forward algorithm,

$$\mathcal{P}(Z_T = s_i | \dot{\mathbf{X}}_{\mathbf{T}}) = \tilde{\alpha}_T(i), \quad \text{for } i \in I_S.$$

Based on these three results, it is possible to formulate a forecasting algorithm in an iterative manner similar to the ones in the previous section. However, a straightforward way is to compute the desired probabilities via matrix multiplication, equal to the usage of matrix notation in regular Markov processes.

Consider $\tilde{\alpha}_T$ as a N -dimensional row vector with $\tilde{\alpha}_T(i)$ as its elements and representing the distribution vector for the possible state position of the hidden process at time T . If we multiply the transition matrix $\mathbb{A}$ on the right side, we derive the distribution vector

of the hidden process at time $T + 1$, verifiable through equation 2.2.3. Repeating this multiplication H -times resolves in the distribution vector of time instance $T + H$,

$$\tilde{\alpha}_T \cdot \mathbb{A}^H = [\mathcal{P}(Z_{T+H} = s_i | \dot{\mathbf{X}}_{\mathbf{T}})]_{i \in S_I}.$$

Applying equation 2.2.3 results in the additional multiplication of the observation matrix $\mathbb{B}$. We introduce the M -dimensional row vector θ denoting the desired distribution vector over the observation space,

$$\theta = [\mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}})]_{k \in S_O} = \tilde{\alpha}_T \cdot \mathbb{A}^H \cdot \mathbb{B}.$$

The forecasting process consists therefore of two steps. First, computing $\tilde{\alpha}_T$ via the adapted forward algorithm and secondly, calculating θ through matrix multiplication.

HMM Forecasting Method

1. Step:
Calculate $\tilde{\alpha}_T$ via the adapted forward algorithm
2. Step:
 $\theta = \tilde{\alpha}_T \cdot \mathbb{A}^H \cdot \mathbb{B}$

return θ

The computational complexity of a single isolated forecast is of the order $O(N^2 \cdot (T + H))$. Even though this computational effort is already viable, the forecasting efficiency can be improved by the following thoughts. Given that in the forecasting step the parameters and thus N are fixed, the two influencing factors are the historic window length T and the forecasting horizon H .

First, discussing the historical factor, recent historic events greatly influence the future outcome, while events longer ago do not bring much additional information. Although using all available data maximizes the available information, the question is how much historic data is truly needed to include all essential information and achieve equally good predictions. Depending strongly on the parameters of the specific HMM, the historic window length T can be reduced to a reasonable length accordingly. In the upcoming chapters of this work, we present an exemplary analysis of the correct choice of T . Furthermore in practice, when new forecasts are generated for each new time instances (for example at time instance T and then at $T + 1$) the previous calculated quantities $\tilde{\alpha}_T$ can be reused for the computation of the new $\tilde{\alpha}_{T+1}$, which cuts down the computational time significantly.⁴

Secondly, turning to the future horizon H , we can assess the quality of the forecast. Generally, the farther ahead in the future the predicting event is, the less precise the forecast it will be. In the HMM setting, this is determined by the characteristics of the transition matrix $\mathbb{A}$. Given that the hidden Markov process fulfills certain conditions⁵,

⁴For scientific reproducibility, this idea is not applied in the experiments of this thesis.

⁵For irreducible, aperiodic and recurrent Markov chains.

the process converges to the stationary distribution $\delta = \delta \cdot \mathbb{A}$. Thus, also the forecast $\tilde{\alpha}_T \cdot \mathbb{A}^H \cdot \mathbb{B}$ converges with increasing H towards a fixed distribution (which is the empirical distribution of the observation process). Implying that with increasing H , the informative value of the forecast decreases and thus, putting an upper bound to the reasonable usage of HMM as a forecasting method.

The predicted probabilities $\mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}})$ contain all information available through the HMM model, and therefore, the resulting distribution vector θ represents the prediction with the maximum amount of information. While in the present thesis, we apply θ directly as a probabilistic forecast, these probabilities can also be utilized as a basis for other types of forecasts. This can be described by the abstract mapping function f using θ as input,

$$f(\theta).$$

Depending on the application, the observation space and objective, f can take on various forms. Here, we present some suitable use cases in the following table 2.2.3, while Gneiting and Katzfuss [10] thoroughly discuss the theoretical reasoning and properties of statistical functionals of probabilistic forecasts.

Function f	Application
$\arg \max$	point forecast for categorical data
$\mathbb{E}$, median	point forecast for numerical data
$\mathbb{V}$, quantiles, confidence interval	measure of uncertainty of the forecast

Table 1.: Overview of functions and their applications.

3. Experiment Design

This chapter provides a detailed explanation on how HMM can be used in the specific use case of forecasting probabilistically the electrical power load of households. This problem is potentially relevant for two parties. Home owners are confronted with an increase in electric demand in the upcoming future due to the electrification of transport (E-mobility) and heating (heat pumps). Using home energy management systems (HEMS) can optimize the operation of the home system via different components (PV electricity production, battery storage management, flexible demand, dynamic pricing) for which good demand predictions are valuable. Secondly for distribution grid providers, accurate forecasts are essential to avoid grid congestion and re-dispatching. HMM models offer here the possibility to apply a simple and efficient agent based forecasting method.

Building on the theory presented in the previous chapter, we discuss a general methodology for applying HMM-based probabilistic forecast and then present the use case including a discussion of the real historic households data and a detailed analysis of the correct parameter setting of the HMM model.

3.1. Methodology

The task is to predict the distribution of the future electric demand of a household only given historical load data. The historical data is represented by a time series with real valued elements in the electric power unit watts[W]. In order to apply discrete HMMs for continuous data, discretization of the data is first needed. Also, we assume training and forecasting as isolated problems and no learning during the forecasting process is allowed. Thus, we only train the model once and use this model for multiple forecasts, although more data may be available in future time steps. The methodology is therefore structured into preprocessing, training, and forecasting.

3.1.1. Preprocessing

In preprocessing, the data is discretized from the space of real numbers $\mathbb{R}$ to the discrete observation space O . Thus, there are three variabilities: the number of observations N , the selection of observations in O , and the mapping from $\mathbb{R}$ to O . We assume N as a model hyperparameter, which we have to optimize. The remaining two factors are determined by the classic discretization principle. The original observation space $\mathbb{R}$ is divided into disjunct intervals called bins, with each bin having one representative value. In the discretization process, each real-valued observation is then replaced by the representative value of the corresponding bin in which it is lying in. Since the choice of the intervals and the representative values can have a significant impact on the models

performance, we propose two different discretization methods, which both will be studied further.

Discretization Type A: Equal-mass bins

The bins are selected so that each bin includes the same number of observations. More precisely, the interval limits of the bins are selected based on the empiric quantiles¹. As representative value, the median of each bin is chosen. Therefore, high frequented areas of the observation space have a higher resolution than low frequented areas. This discretization method benefits the stochastic design of the HMM model, because the resulting transition and observation probabilities will be balanced due to the balanced occurrences of each discrete observation.

Discretization Type B: Equidistant Bins

The original observation space $\mathbb{R}$ is discretized into bins with equal ranges (like eg. 0-100 Watts, 100-200 Watts, 200-300 Watts, etc.). The representative value of each bin is then the middle point of the bin's intervals. This is the standard discretization method and often desirable from an applied perspective.

3.1.2. Training

The application of HMM to model a one-dimensional signal like electricity demand is designed so that the signal is represented by the observation process $(X_t)_{t \in \{1, \dots, T\}}$ while the hidden process $(Z_t)_{t \in \{1, \dots, T\}}$ is unknown. In our case, we can measure the total amount of electricity demand of a household but do not know the underlying process. This process is influenced by the inhabitants, the electric devices and other factors like weather or season. The advantage of HMM is that this hidden process does not need to be specified. During the training phase, the optimal parameters of the model are generated without deeper knowledge of the process. Via decoding algorithms, the most probable hidden state sequences can be identified and interpreted in a post-training analysis. For example, low energy emitting states can be connected to night time hours or times when nobody is at home. The only condition for the hidden state process is to satisfy the Markov conditions which are quite strong.

The hyperparameters of our HMM model are the type of discretization, the size of the observation space M and the number of hidden states N . We will conduct a detailed hyperparameter analysis explained later on. During the training phase, the parameters λ of the HMM are trained with the Baum-Welch algorithm given the specific hyper parameter setting and the historic observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$. The Baum-Welch algorithm is run using the fixed settings of random initial parameters and a maximum number of iterations set to 100 which has been shown to be sufficient. Since the training algorithm relies strongly on the empiric occurrences of the observations, in the selection of the training data data-specific bias should be avoided. In the case of household's electric power load, it is known that there are strong daytime and seasonal differences. Therefore, only data including full days and years should be included.

¹For example, when the number of discrete observations is fixed to N , the n 'th bin has as left interval limit the $(n-1)/N$ 'th empiric quantile and as the right limit the n/N 'th empiric quantile.

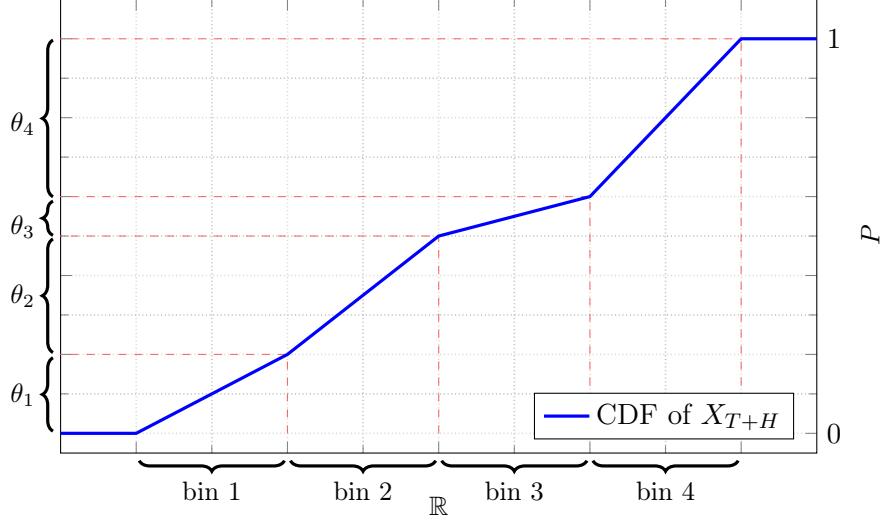


Figure 3.: Continuation of the distribution vector θ with four exemplary bins.

3.1.3. Forecasting and Evaluation

Given a trained HMM with parameters λ , we can conduct probabilistic forecasts as explained in 2.2.3. The two relevant parameters for forecasting are the historic window length T and the future horizon H . For the first parameter T we will use a fixed length and conduct a sensitivity analysis later to verify the choice. The future horizon H is set to 1, thus, it will be a one-step-ahead prediction. Even though it is the most simple forecast, we can derive a lot of information about the quality of the forecast because predictions further ahead build upon the same information as the one-step-ahead prediction.

The resulting prediction from the HMM forecast is the discrete distribution vector θ over the discretized observation space O , however, the original problem requested a distribution forecast in the continuous space $\mathbb{R}$. For this reason, the discrete distribution is re-converted to a continuous CDF with following methodology. Recalling that the distribution vector θ represents the probabilities of the corresponding bins, we presume that values inside each bin have equal densities. Assuming the bins are sorted ascending, the probability of X_{T+H} being smaller than the right interval limit of bin k is the sum of the probabilities of all bins l lower and equal to bin k ,

$$\mathcal{P}(X_{T+H} \leq \text{right interval limit of bin } k) = \sum_{l=1}^k \theta_l.$$

This leads to a piecewise-linear CDF, where the above defined supporting points are connected by straight lines as shown exemplary in graph 3.

Given the continuous CDF as predicted distribution, the forecast is then evaluated versus the true realization via the discrete CRPS. To achieve statistical valid results, this procedure is repeated for each time instances of a given test set and the mean-CRPS is taken.

3.2. Data

The data source for the experiments presented in this work is the WPuQ dataset collected by Schlemminger, Ohrdes, Schneider, and Knoop [20]. This dataset contains the electrical power load data of 38 single-family houses in northern Germany with the additional information of PV-production, separated heat pump load, weather data, household-specific information and distribution grid measurements. The electrical power was measured over two and a half years during the period of 2018 to 2020 in 10 second intervals. Missing measurements are either labeled as missing value or in cases of missing data gaps smaller than one day linearly interpolated. The authors also provide aggregated data over longer time intervals (1, 15 and 60 minutes) in which the mean power values of the original measurements are taken.

For this thesis, the experiments are conducted on 5 out of the 38 households of the WPuQ. These 5 households were randomly selected with the condition that they do not have PV production and only a minimal number of values are interpolated ($<1\%$). The selection of households are described in table 3.2 and denoted with HH1-HH5 in this paper.

Selected Household	WPuQ household ID
HH1	SFH3
HH2	SFH4
HH3	SFH9
HH4	SFH12
HH5	SFH14

Table 2.: Mapping between selected households and WPuQ household IDs

For each of these households, one single time series is taken, namely the total² active power load, in 15 minutes resolution. The 15 minutes resolution is chosen because this is the current standard resolution of electricity data in most European and especially Austrian energy systems (like eg. EPEX spot market, smart meter data transmission,...). In addition, each of the time series is normalized by dividing the original time series with its maximum elements. Thus, each value lies in the range of 0 and 1³. We split the data into a training, validation and test datasets. On the training data set, the forecasting models will be trained, the validation data set is used for evaluating different HMM models and finding the optimal hyperparameter setting, and the test set is applied for benchmarking. To avoid seasonal biases, the training data set includes all values of year 2019, which is the first full year with available data. The second fully available year, 2020, is used for validation and testing. Again, to avoid seasonal biases in the evaluation, every odd-numbered month of 2020 (January, March, May,...) is assigned for validation while every even-numbered month of 2020 (February, April, June,...) is assigned to the test

²Total is in the context of the sum of all three measured phases of the households electric circuit.

³No negative values are feasible since households with PV production are excluded.

data set.

To emphasize the importance of unbiased data sets, Figure 4 demonstrates the time series pattern and differences concerning day times and seasonality. We refer to the paper of Schlemminger, Ohrdes, Schneider, and Knoop [20] for a in-depth analysis of the data characteristics of households electrical power demand.

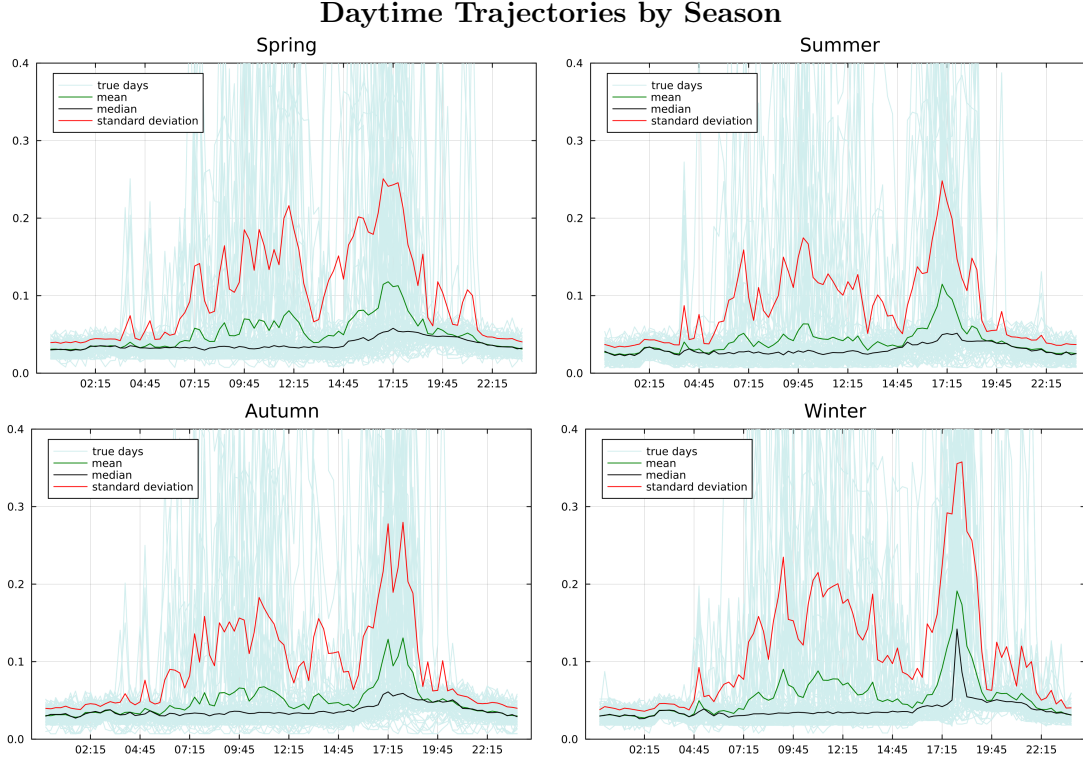


Figure 4.: In these figures, the daytime trajectories of all days for each season are displayed in gray with statistics like the mean, median, and mean plus standard deviation. All four seasons show similar load profiles with obvious differences during day and night time. Event hough the profiles of different seasons have a similar shape, the variability of the data is significantly greater during the winter months then during the summer months.

3.3. Experiments

The general goal of the experiments is to find the best HMM forecasting models for one-step-ahead predictions. For this reason, not only the optimal parameter settings are trained, but also the right choice of hyperparameter has to be studied. There are four different hyperparameters that result from the methodology: the discretization method, the number of hidden states N , the number of observations M , and the historic window

length for forecasting T . For each discretization method, the optimal settings of N and M are elaborated with a hyperparameter analysis, while the correct choice of T is verified afterwards with a sensitivity analysis. All experiments are conducted for each household and discretization type independently, resulting in two final HMM forecasting models per household.

Hyperparameter Analysis

In the hyperparameter analysis, we conduct a grid search to find the optimal settings of the number of hidden states N and observations M for each household and discretization type A and B . For each hyperparameter setting, the optimal parameters of the HMM model are trained via the Baum-Welch algorithm on the test data set of the household. The models are then evaluated using the validation data set and computing the mean-CRPS of the one-step-ahead prediction for each validation time instance. Hereby, we fix the historic window length H to be 100 which is a length of slightly more than one day considering the data time resolution of 15 minutes. The tested values for the grid search of N and M additional to the other fixed hyperparameters are summarized in table 3. All combinations of these hyperparameter settings are evaluated resulting in the training and testing of 40 models per household and discretization type.

Hyperparameter	Values
N	10, 20, 30, 40, 50, 60, 70, 80, 100
M	25, 50, 100, 200
H	100

Table 3.: Overview of hyperparameter settings used in the hyperparameter analysis.

Sensitivity Analysis

In the sensitivity analysis, we investigate the effect of change in the historic window length H on the models forecasting performance. Hereby, we aim to verify the choice of H of the hyperparameter anal but also get insights of the HMM forecasting behavior. Since the historic window length H is only relevant during the prediction step and not during training, the models do not have to be retrained. Building up onto the previous results, we use the best performing models for each household and discretization type and evaluate the mean-CRPS with variable values of H . This evaluation is conducted once again on the validation data set and the result is analyzed visually. The tested values are described in table 4.

Hyperparameter	Values
N	fixed to optimal
M	fixed to optimal
H	1, 2, 3, 5, 10, 25, 50, 100, 200

Table 4.: Overview of hyperparameter settings used in the sensitivity analysis.

4. Experiment Results

This chapter presents the results of experiments concerning the application of HMM forecasting models for the prediction of households electric power loads. Sections 4.1 and 4.2 cover the outcome of the hyperparameter and sensitivity analysis, while in the third section, we analyze some properties of the resulting optimal models. This method provides further insights of the characteristics of HMM forecasting models and can also be employed for other future applications.

4.1. Hyperparameter Analysis

For each household and both discretization types, 36 HMM models with different hyperparameter settings were trained and evaluated. The first representative figure 5 visualizes the results of the hyperparameter analysis of HH1. In appendix A, the reader finds the corresponding figures for HH2-HH5 as well as a complete table with the exact numerical results of the hyperparameter analysis. A summary of the numeric results is given by table 5, which highlights the hyperparameter settings of the optimal models with its corresponding mean-CRPS values. From the hyperparameter analysis figures, the following conclusions are drawn.

First, the number of observations have a significant impact on the models performance. The choice of discretizing the observation space in only 25 observations leads in almost every setting to the worst results, but also a high resolution of 200 observations influences the performance of the models badly. The number of observations of 50 and 100 achieves the most consistent results with 100 observations being the best setup for all households, except for HH3 with discretization type A.

Further, the optimal number of hidden states differs for each household. However, a major

Household	N	M	CRPS
HH1	40	100	0.8232
HH2	30	100	0.6846
HH3	50	50	1.3658
HH4	60	100	0.9505
HH5	40	100	0.8219

(a) Discretization Type A

Household	N	M	CRPS
HH1	80	100	0.8259
HH2	40	100	0.6874
HH3	50	100	1.3587
HH4	50	100	0.9157
HH5	80	100	0.8213

(b) Discretization Type B

Table 5.: The hyperparameter setting of the optimal models for each household

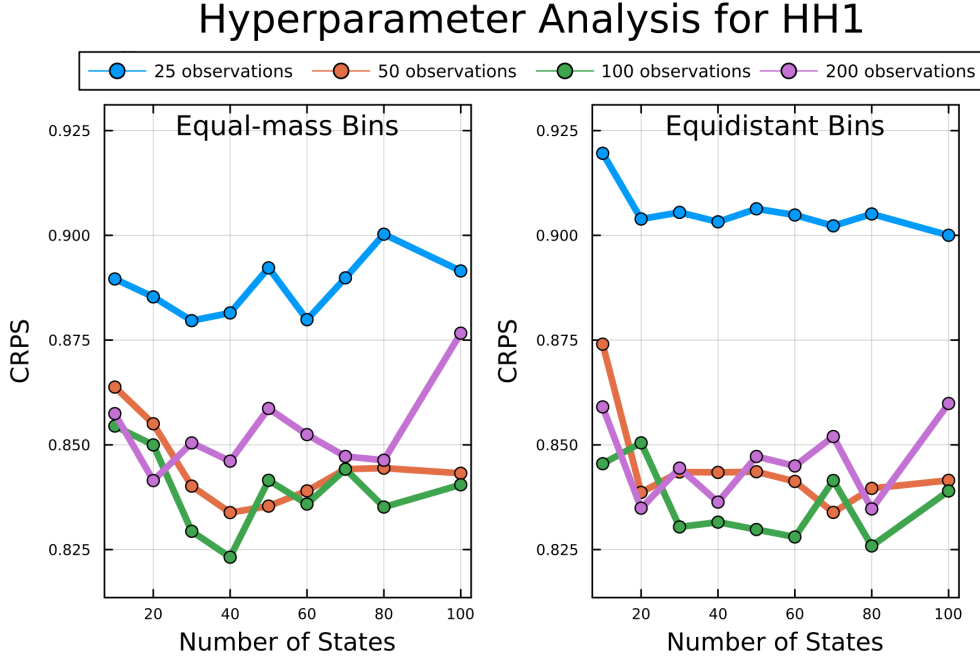


Figure 5.: Representative graph of the hyperparameter analysis of HH1. The left graph displays the results for discretization type *A* with equal-mass bins, while the graph on the right side shows the results for discretization type *B* with equidistant bins. The optimal models are chosen by the minimal mean-CRPS value, which is for discretization type *A* the model with 100 observations and 40 hidden states and for discretization type *B* the model with 100 observations and 80 hidden states.

result of this analysis is that models of discretization type *A* reach their optimal setting with lower a lower number of hidden states than models of discretization type *B*. This effect is explained due to the design of discretization. The discretization of equal-mass bins favors the stochastic characteristic of the HMM models because it leads to a balanced weighting of all observation and thus the underlying hidden states.

Comparing the performance of the discretization methods, we observe for all households except HH4 that both discretization methods achieve similar optimal mean-CRPS values. However, in general, the optimal models discretized with equal sized bins are consistently overall optimal for each household. Summarizing, it can be concluded that discretization type *A* leads to smaller and more efficient models, while models with discretization type *B* are more robust in the CRPS evaluation.

4.2. Sensitivity Analysis

The sensitivity analysis investigates the convergence of the prediction performance conditioned by increasing historic window length. The experiments were conducted for each household and discretization type with the optimal models hyperparameter setting summarized in 5. The representative figure 6 visualizes the results of the sensitivity analysis for HH1, while the corresponding figures for HH2-HH5 are in appendix B.

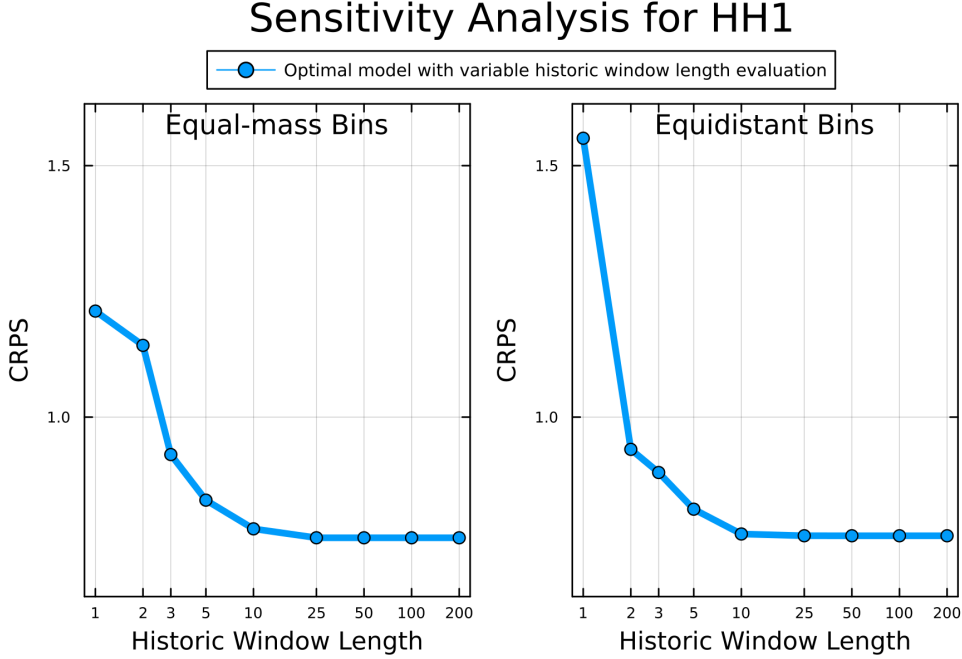


Figure 6.: Representative figure of the result of the sensitivity analysis for HH1.

The results of the sensitivity analysis support the hypothesis that with increasing the historic window length and thus increasing available information, the performance of the forecasts improve. All household show a similar pattern of converging mean-CRPS to a lower limit. This limit is in most cases reached at $H = 10$, for HH2 already at $H = 5$ and only in the case of HH1 with discretization in equal-mass bins at $H = 25$. There are no significant differences between the discretization types, even though they differ greatly in the number of hidden states, like e.g. in HH1.

In addition, it verifies that the choice of H during the hyperparameter analysis was well taken. For further experiments in the future, the historic window length can be chosen even lower to improve computational evaluation times.

Lastly, considering the converging point to be around 10, it can be said that only historic data up to two and a half hours ago adds beneficial information to the HMM forecasting model. That also implies that the model does not detect recognizable behavior and patterns before this point, inferring that HMM forecasting models should only be applied for short-time pattern forecasting.

4.3. Model Analysis

DIESE SECTION IST EIN PROVISORIUM:

This section investigates the behavior of the optimal models. In specific, we conduct different PIT histograms to assess the calibration of our forecasts and study the convergence of the predicted distributions with increasing forecasting horizon.

While the mean-CRPS, as obtained in the numerical results of this chapter, considers calibration, it is also influenced by the sharpness of the predicted distribution. To get more insight into the models predictive power and behavior, we generate the PIT histograms for the optimal models. This provides a visual judgment on the calibration of the models. There exist many in-depth studies how to conduct a numerical decomposition of the CRPS into a calibration and sharpness part [12]. The PITs were generated over the validation data set for each one-step-ahead prediction. Figure 7 shows the PIT histograms for both optimal models of HH1, the figures of the other households can be found in appendix C.1.

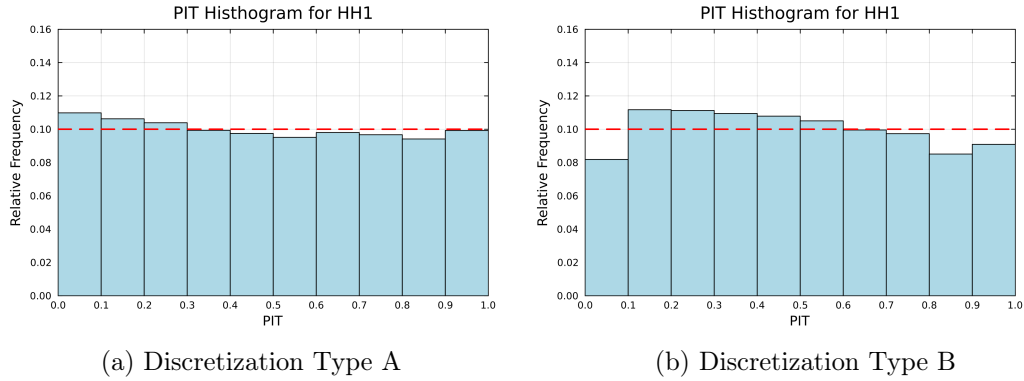


Figure 7.: Representative PIT histograms for HH1

From these PIT histograms, it is visible that throughout all households the model with discretization type *B* have a worse calibration than the ones with discretization type *A*. This effect can be seen especially strong for HH4. Implying that the reliability of the prediction increases if the discretization with equal-mass bins is carried out. Considering the similar values of the mean-CRPS in the households for different discretization methods, we can assume that discretization type *B* models compensate the lack of calibration with a greater sharpness explainable through its bigger model size (number of hidden states). In general, a better calibrated model should always be preferred recalling the major principle of probabilistic forecasting.

The last study on our model investigates the model behavior for a forecasting horizon larger than one, as applied until now. We are not interested in the numerical performance of the models, however, want to know how far into the future our HMM models can be reasonably used. For this reason, we investigate the convergence of the predicted distribution θ with increasing forecasting horizon H . Convergence means that the predicted distribution does not change later than a certain point in the future. To illustrate the idea of convergence, figure 5 shows a typical situation of a multi-step-ahead

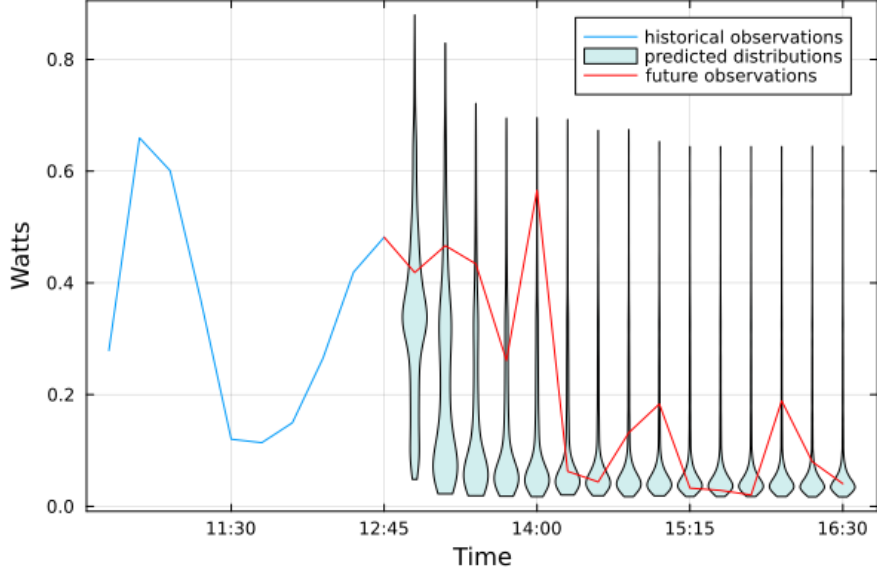


Figure 8.: Example of a typical situation of a multi-step-ahead distribution forecast. Given the historic observation, a distribution forecast is computed for each time instance in the future illustrated by the multiple violin plots. Additionally, the true realization of the future time instances were added to the plot afterwards. Examining the shape of the single distributions, the convergence is visible. From time instance 14:00 onward, the violins do not change significantly and from 15:15 onward, they are visually not differentiable.

distributional forecast. Generally speaking, the convergence of a distributional forecast can be meaningful in certain application. However, with data having daytime patterns as in our case, the convergence of the forecast only shows the limitation of the HMM model which is induced by the stationary properties of Markov models.

Formulating the idea of convergence into a mathematical way, the stochastic vector θ lies on a hyperplane in the N -dimensional real space. We denote θ_h as the predicted distribution h steps into the future, for $h \in \{1, \dots, H\}$ and consider h' as the *convergence point* if it is the first instance inside the epsilon-neighbourhood around the empiric distribution $\delta \times \mathbb{B}$ of the model with the stationary distribution δ of $\mathbb{A}$, thus

$$\|\theta_h - (\delta \times \mathbb{B})\|_1 < \epsilon \quad \text{for all } h > h'.$$

The L^1 metric is chosen, so that the choice of epsilon defines the absolute difference between the stochastic vectors. For example with $\epsilon = 0.01$, convergence is achieved when the predicted distribution vectors only differ up to 1% in total. We experiment with two different epsilon settings, first with $\epsilon = 0.1$ for a rough convergence of the distributional shape and $\epsilon = 0.01$ for a detailed convergence.

The representative figures 10 and 9 display the histograms for all convergence points evaluated over every instance of the validation data set and each optimal model of HH1.

4. Experiment Results

The histograms of the other households are found in the appendix sections C.2 and C.3. Trivially, the convergence to the 10% epsilon-neighborhood is reached earlier than in the 1% epsilon-neighborhood. Referring to the rougher convergence, a lot of forecasts do not significantly change its shape around 5 steps into the future, most forecasts come when before 10 steps, and only a few forecasts do change significantly their shape only later than 10 (like eg. in HH1.). With a finer convergence of $\epsilon = 0.01$, these boundaries shift to the right. In this case, there are also great differences between the household. Especially HH1 and HH5 require more steps to reach their stationary distributions.

These converging points serve as an absolute limit for H for these HMM forecasting to be applied in a reasonable way, since they converge to the empirical distribution and do not utilize beneficial information anymore.

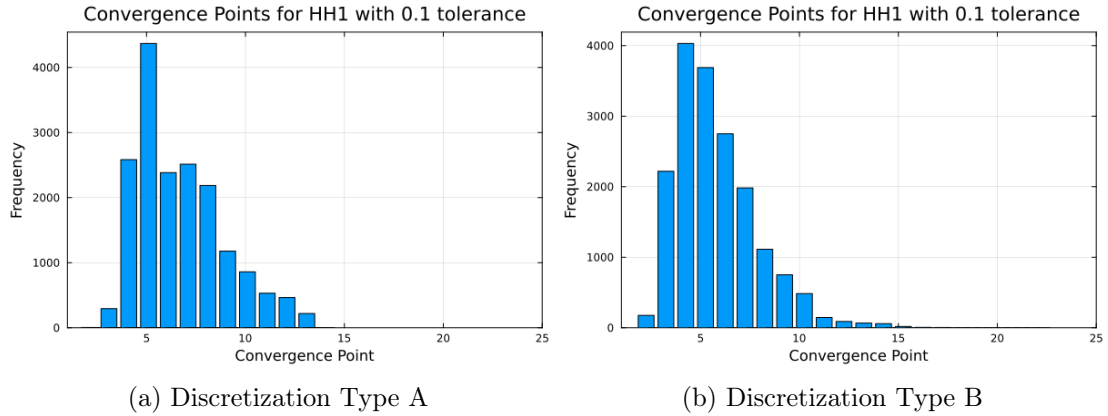


Figure 9.: Convergence point histogram for $\epsilon = 0.1$ and HH1

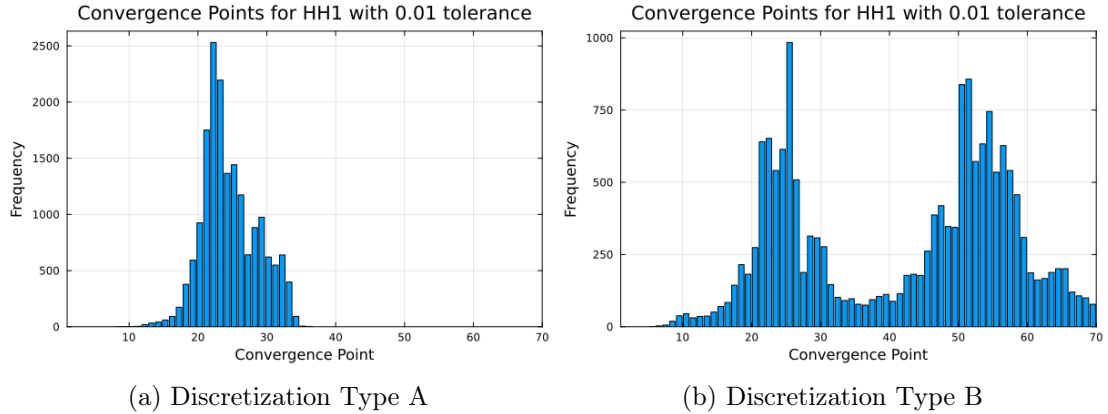


Figure 10.: Convergence point histogram for $\epsilon = 0.1$ and HH1

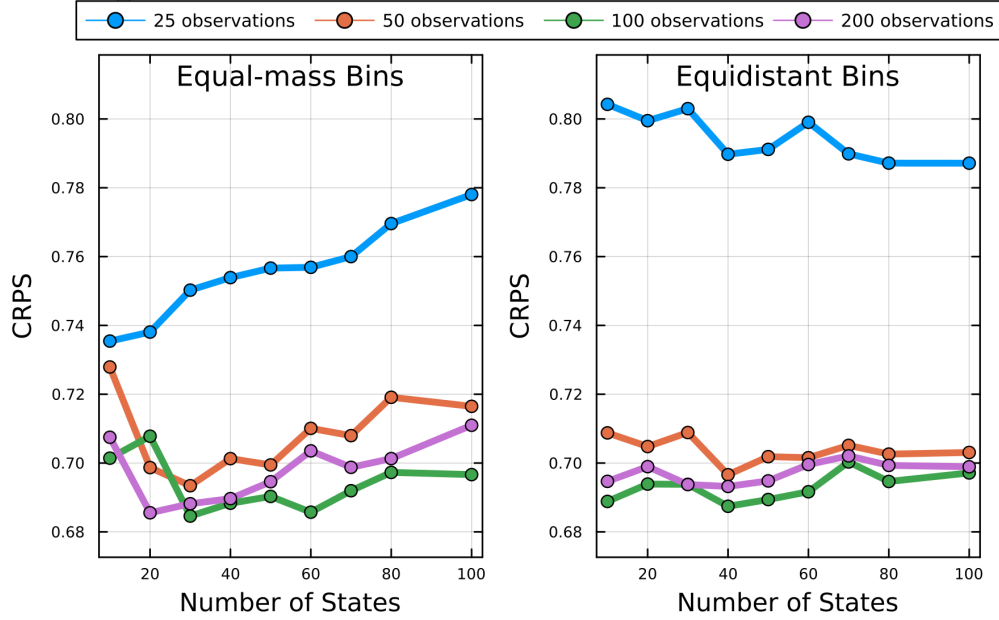
Bibliography

- [1] L. E. Baum. „An Inequality and Associated Maximization Technique in Statistical Estimation for Probabilistic Functions of Markov Process“. In: *Inequalities* 3 (1972), pp. 1–8.
- [2] L. E. Baum and T. Petrie. „Statistical Inference for Probabilistic Functions of Finite State Markov Chains“. In: *The Annals of Mathematical Statistics* 37.6 (1966), pp. 1554–1563.
- [3] L. Botman, J. Lago, T. Becker, K. Vanthournout, and B. D. Moor. „A global probabilistic approach for short-term forecasting of individual households electricity consumption“. In: *Applied Energy* 382 (2025), p. 125168.
- [4] M. Brand, N. Oliver, and A. Pentland. „Coupled hidden Markov models for complex action recognition“. In: *Proceedings of IEEE Computer Society Conference on Computer Vision and Pattern Recognition*. 1997. DOI: [10.1109/CVPR.1997.609450](https://doi.org/10.1109/CVPR.1997.609450).
- [5] A. Brockwell. „Universal Residuals: A Multivariate Transformation“. In: *Statistics and probability letters* 77 (2007), pp. 1473–1478. DOI: [10.1016/j.spl.2007.02.008](https://doi.org/10.1016/j.spl.2007.02.008).
- [6] H. Cloke and F. Pappenberger. „Ensemble Flood Forecasting: A Review“. In: *Journal of Hydrology* 375 (2009), pp. 613–626. DOI: [10.1016/j.jhydrol.2009.06.005](https://doi.org/10.1016/j.jhydrol.2009.06.005).
- [7] A. P. Dempster, N. M. Laird, and D. B. Rubin. „Maximum Likelihood from Incomplete Data Via the EM Algorithm“. In: *Journal of the Royal Statistical Society: Series B (Methodological)* 39.1 (1977), pp. 1–22. DOI: [10.1111/j.2517-6161.1977.tb01600.x](https://doi.org/10.1111/j.2517-6161.1977.tb01600.x). eprint: https://academic.oup.com/jrsssb/article-pdf/39/1/1/49117094/jrsssb_39_1_1.pdf.
- [8] C. Erlwein, F. E. Benth, and R. Mamon. „HMM filtering and parameter estimation of an electricity spot price model“. In: *Energy Economics* 32.5 (2010), pp. 1034–1043. DOI: <https://doi.org/10.1016/j.eneco.2010.01.005>.
- [9] K. Ghasvarian Jahromi, D. Gharavian, and H. R. Mahdiani. „Wind power prediction based on wind speed forecast using hidden Markov model“. In: *Journal of Forecasting* 42.1 (2023), pp. 101–123. DOI: <https://doi.org/10.1002/for.2889>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/for.2889>.
- [10] T. Gneiting and M. Katzfuss. „Probabilistic Forecasting“. In: *Annual Review of Statistics and Its Application* 1. Volume 1, 2014 (2014), pp. 125–151. DOI: <https://doi.org/10.1146/annurev-statistics-062713-085831>.
- [11] J. J. J. Groen, R. Paap, and F. Ravazzolo. „Real-Time Inflation Forecasting in a Changing World“. In: *Journal of Business & Economic Statistics* 31.1 (2013), pp. 29–44. DOI: [10.1080/07350015.2012.727718](https://doi.org/10.1080/07350015.2012.727718).

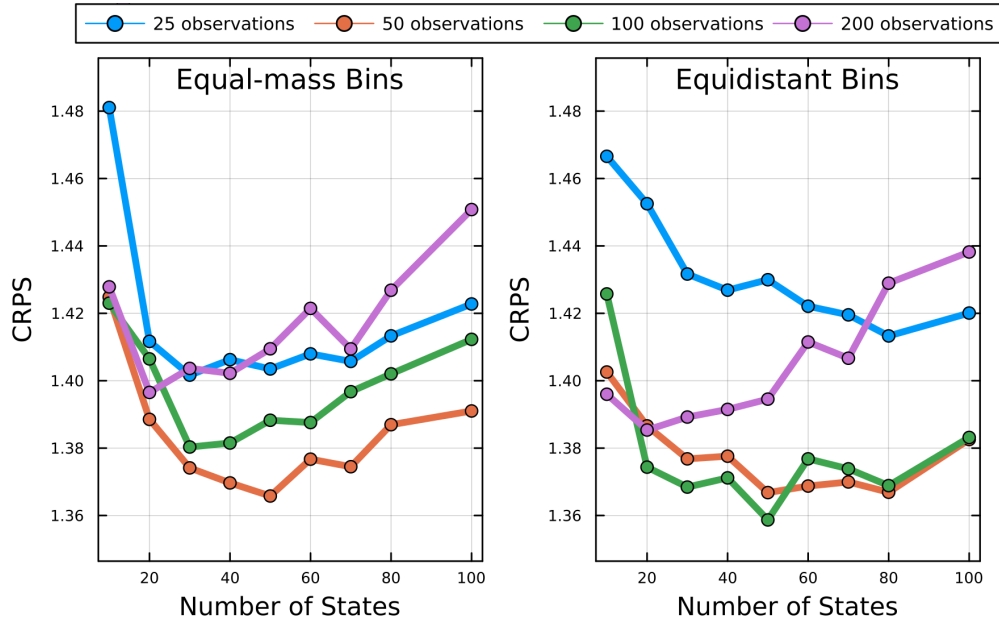
- [12] H. Hersbach. „Decomposition of the Continuous Ranked Probability Score for Ensemble Prediction Systems“. In: *Weather and Forecasting* 15.5 (2000), pp. 559–570. DOI: [10.1175/1520-0434\(2000\)015<0559:D0TCRP>2.0.CO;2](https://doi.org/10.1175/1520-0434(2000)015<0559:D0TCRP>2.0.CO;2).
- [13] T. Hong, P. Pinson, S. Fan, H. Zareipour, A. Troccoli, and R. J. Hyndman. „Probabilistic energy forecasting: Global Energy Forecasting Competition 2014 and beyond“. In: *International Journal of Forecasting* 32.3 (2016), pp. 896–913.
- [14] T. H. Jordan. „The Value, Protocols, and Scientific Ethics of Earthquake Forecasting“. In: *EGU General Assembly Conference Abstracts*. 2013, EGU2013-12789.
- [15] D. Kulp, D. Haussler, M. G. Reese, and F. H. Eeckman. „A Generalized Hidden Markov Model for the Recognition of Human Genes in DNA“. In: 1996. URL: <https://www.scopus.com/inward/record.uri?eid=2-s2.0-0030333286&partnerID=40&md5=831af18283d8d9dcdd265177d2bbe2e4>.
- [16] S. Makonin, F. Popowich, I. V. Bajić, B. Gill, and L. Bartram. „Exploiting HMM Sparsity to Perform Online Real-Time Nonintrusive Load Monitoring“. In: *IEEE Transactions on Smart Grid* 7.6 (2016), pp. 2575–2585. DOI: [10.1109/TSG.2015.2494592](https://doi.org/10.1109/TSG.2015.2494592).
- [17] A. H. Murphy and R. L. Winkler. „Probability Forecasting in Meterology“. In: *Journal of the American Statistical Association* 79.387 (1984), pp. 489–500.
- [18] J. Nowotarski and R. Weron. „Recent advances in electricity price forecasting: A review of probabilistic forecasting“. In: *Renewable and Sustainable Energy Reviews* 81 (2018), pp. 1548–1568. DOI: <https://doi.org/10.1016/j.rser.2017.05.234>.
- [19] L. Rabiner. „A tutorial on hidden Markov models and selected applications in speech recognition“. In: *Proceedings of the IEEE* 77.2 (1989), pp. 257–286.
- [20] M. Schlemminger, T. Ohrdes, E. Schneider, and M. Knoop. „Dataset on electrical single-family house and heat pump load profiles in Germany“. In: *Scientific Data* 9 (2022), p. 56. DOI: [10.1038/s41597-022-01156-1](https://doi.org/10.1038/s41597-022-01156-1).
- [21] C. Strähl and J. Ziegel. „Cross-calibration of probabilistic forecasts“. In: 11.1 (2017), pp. 608–639. DOI: [10.1214/17-EJS1244](https://doi.org/10.1214/17-EJS1244).
- [22] V. Ter-Hovhannisyan, A. Lomsadze, Y. O. Chernoff, and M. Borodovsky. „Gene prediction in novel fungal genomes using an ab initio algorithm with unsupervised training“. In: *Genome Research* 18.12 (2008), pp. 1979–1990. DOI: [10.1101/gr.081612.108](https://doi.org/10.1101/gr.081612.108).
- [23] R. Tian and G. Shen. „Predictive power of Markovian models: Evidence from US recession forecasting“. In: *Journal of Forecasting* 38.6 (2019), pp. 525–551. DOI: <https://doi.org/10.1002/for.2579>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/for.2579>.

A. Results Hyperparameter Analysis

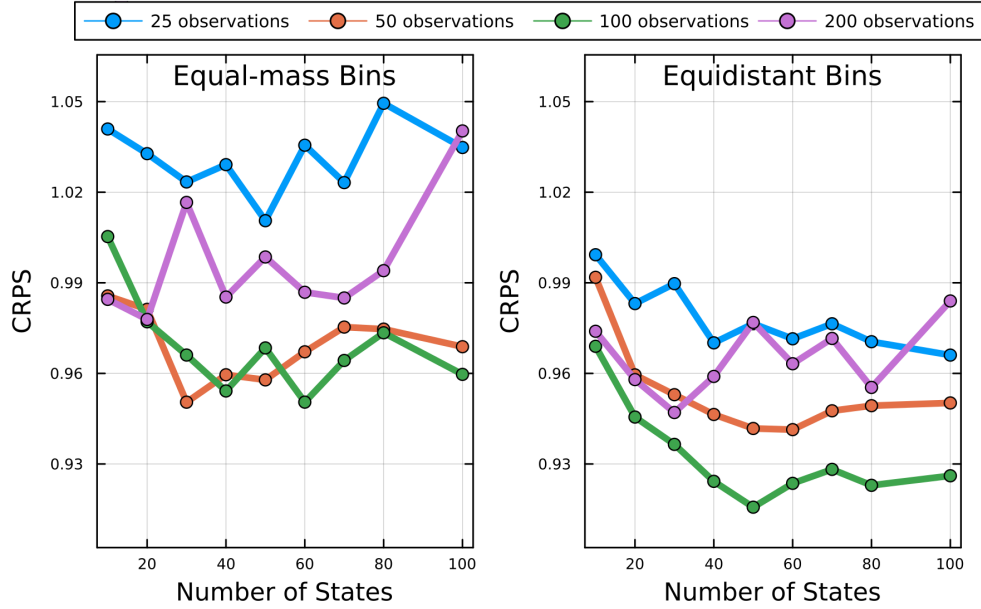
Hyperparameter Analysis for HH2



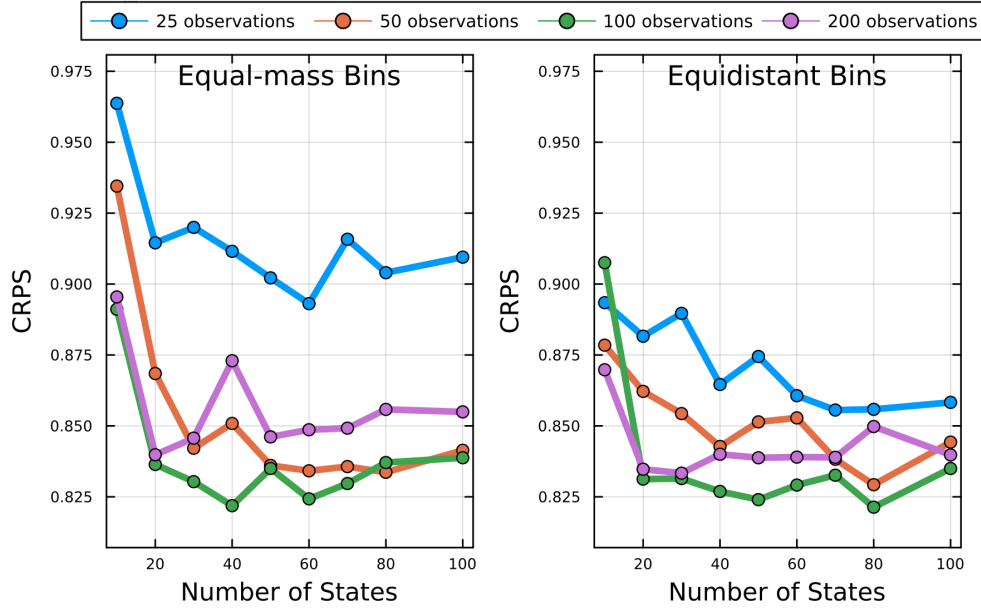
Hyperparameter Analysis for HH3



Hyperparameter Analysis for HH4



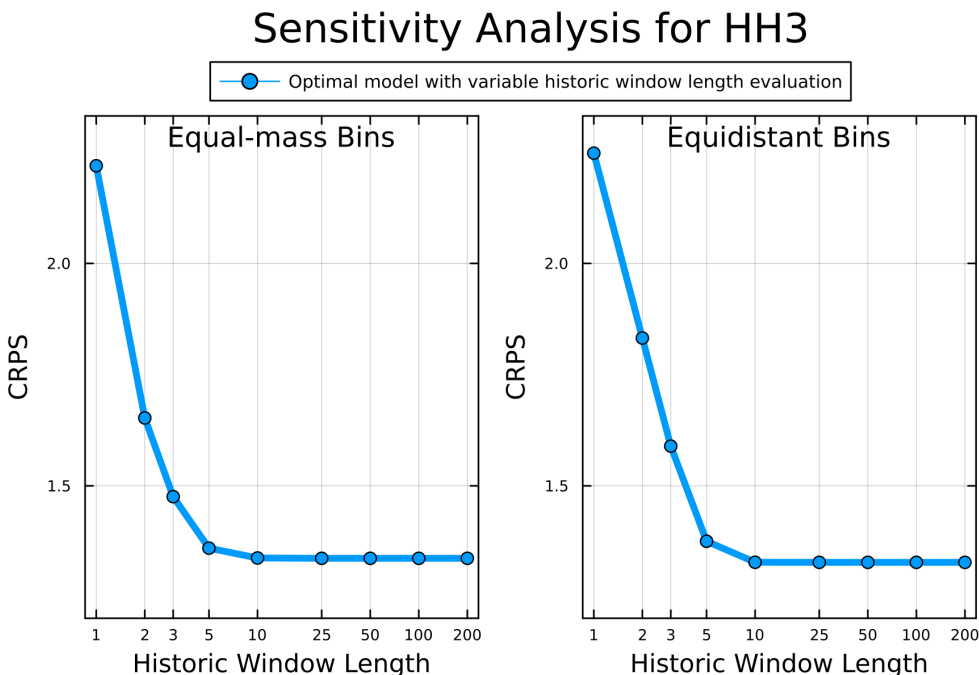
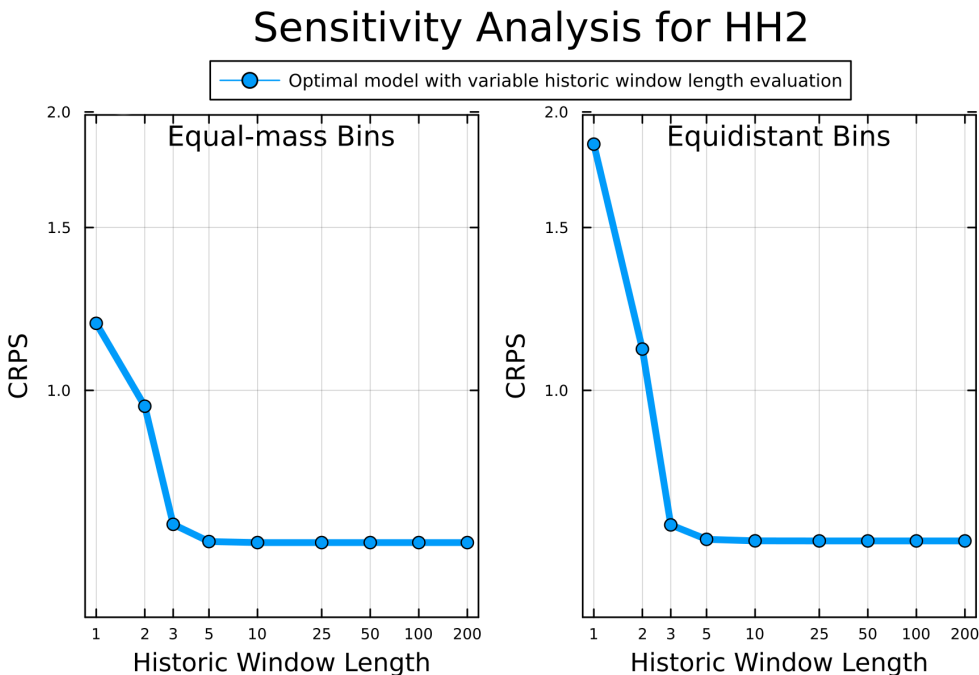
Hyperparameter Analysis for HH5



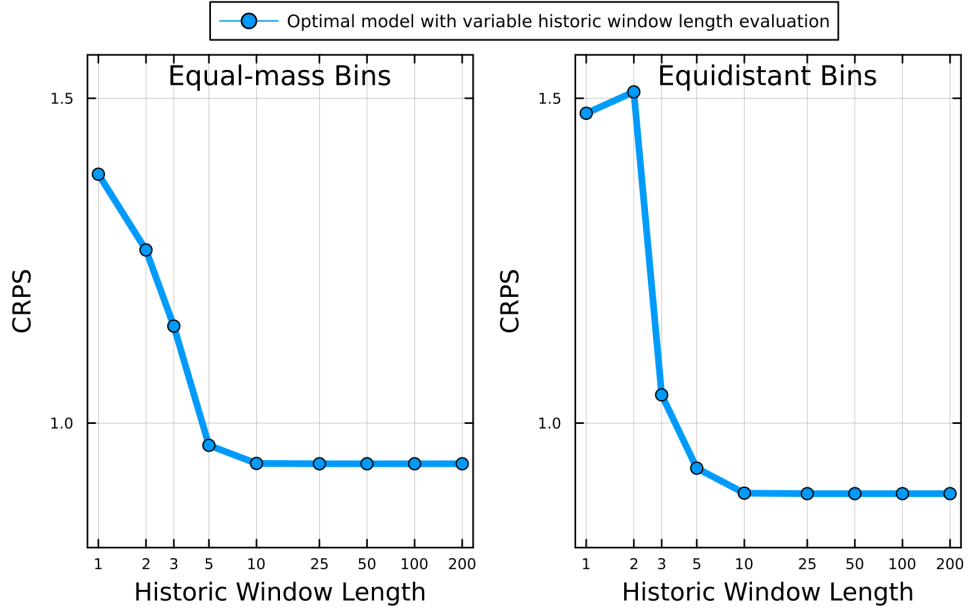
A. Results Hyperparameter Analysis

HH	Discr. Type	M\N	10	20	30	40	50	60	70	80	100
HH1	A	25	0,8896	0,8853	0,8796	0,8815	0,8922	0,8799	0,8899	0,9003	0,8915
		50	0,8638	0,8550	0,8402	0,8338	0,8354	0,8390	0,8442	0,8444	0,8432
		100	0,8545	0,8500	0,8294	0,8232	0,8415	0,8359	0,8442	0,8352	0,8405
		200	0,8574	0,8415	0,8505	0,8461	0,8587	0,8524	0,8472	0,8463	0,8766
	B	25	0,9196	0,9039	0,9055	0,9032	0,9063	0,9049	0,9023	0,9051	0,9000
		50	0,8740	0,8387	0,8435	0,8434	0,8436	0,8413	0,8339	0,8396	0,8415
		100	0,8455	0,8505	0,8304	0,8316	0,8298	0,8280	0,8415	0,8259	0,8390
		200	0,8591	0,8349	0,8444	0,8363	0,8472	0,8450	0,8520	0,8347	0,8599
HH2	A	25	0,7354	0,7381	0,7503	0,7539	0,7567	0,7569	0,76	0,7696	0,778
		50	0,7279	0,6987	0,6934	0,7013	0,6995	0,7101	0,708	0,7191	0,7165
		100	0,7014	0,7078	0,6846	0,6883	0,6903	0,6857	0,692	0,6973	0,6966
		200	0,7075	0,6855	0,6882	0,6897	0,6946	0,7035	0,6988	0,7013	0,711
	B	25	0,8042	0,7995	0,803	0,7897	0,7912	0,799	0,7899	0,7872	0,7871
		50	0,7088	0,7048	0,7089	0,6966	0,7019	0,7016	0,7051	0,7026	0,7031
		100	0,6889	0,6939	0,6938	0,6874	0,6894	0,6917	0,7003	0,6946	0,6971
		200	0,6947	0,699	0,6938	0,6932	0,6948	0,6996	0,7021	0,6993	0,6989
HH3	A	25	1,4811	1,4117	1,4017	1,4062	1,4035	1,408	1,4057	1,4133	1,4228
		50	1,4248	1,3885	1,3741	1,3697	1,3658	1,3767	1,3745	1,387	1,391
		100	1,423	1,4065	1,3803	1,3815	1,3883	1,3876	1,3967	1,402	1,4123
		200	1,4279	1,3965	1,4036	1,4022	1,4095	1,4215	1,4094	1,4269	1,4508
	B	25	1,4666	1,4525	1,4317	1,4269	1,43	1,4221	1,4195	1,4133	1,4201
		50	1,4026	1,3866	1,3768	1,3776	1,3668	1,3687	1,37	1,3669	1,3826
		100	1,4257	1,3744	1,3684	1,3712	1,3587	1,3768	1,3739	1,3689	1,3832
		200	1,396	1,3854	1,3892	1,3915	1,3945	1,4115	1,4067	1,4289	1,4382
HH4	A	25	1,0409	1,0328	1,0234	1,0291	1,0106	1,0356	1,0232	1,0494	1,0348
		50	0,9857	0,9812	0,9506	0,9595	0,9578	0,9672	0,9753	0,9747	0,9688
		100	1,0053	0,9771	0,9661	0,9542	0,9684	0,9505	0,9643	0,9734	0,9597
		200	0,9845	0,9779	1,0166	0,9853	0,9986	0,9869	0,985	0,994	1,0403
	B	25	0,9993	0,9831	0,9897	0,9701	0,9765	0,9715	0,9764	0,9705	0,9661
		50	0,9918	0,9595	0,9529	0,9464	0,9418	0,9414	0,9476	0,9493	0,9502
		100	0,969	0,9455	0,9365	0,9242	0,9157	0,9236	0,9282	0,9229	0,9261
		200	0,9739	0,9579	0,947	0,959	0,9769	0,9632	0,9716	0,9553	0,984
HH5	A	25	0,9637	0,9145	0,92	0,9115	0,9022	0,8931	0,9158	0,904	0,9095
		50	0,9345	0,8685	0,8421	0,8509	0,8361	0,8342	0,8357	0,8336	0,8414
		100	0,8911	0,8364	0,8303	0,8219	0,835	0,8243	0,8297	0,8371	0,8388
		200	0,8955	0,8398	0,8457	0,873	0,8461	0,8487	0,8492	0,8558	0,8549
	B	25	0,8934	0,8816	0,8897	0,8646	0,8745	0,8606	0,8556	0,8558	0,8583
		50	0,8785	0,8622	0,8543	0,8427	0,8514	0,8528	0,8382	0,8293	0,8443
		100	0,9075	0,8312	0,8314	0,8269	0,824	0,8291	0,8326	0,8213	0,835
		200	0,8698	0,8347	0,8333	0,84	0,8388	0,839	0,8389	0,8498	0,8398

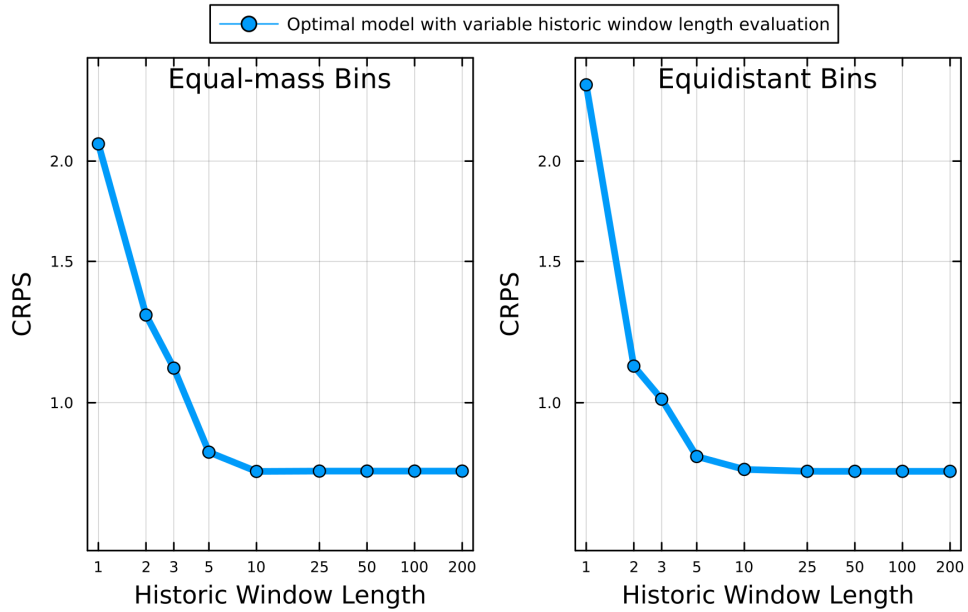
B. Results Sensitivity Analysis



Sensitivity Analysis for HH4

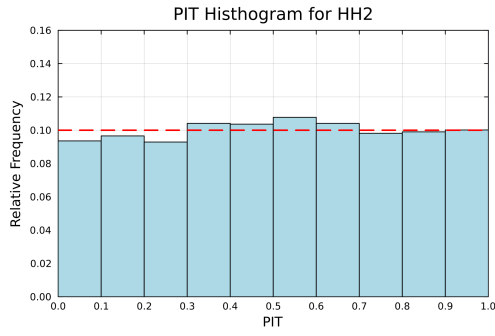


Sensitivity Analysis for HH5

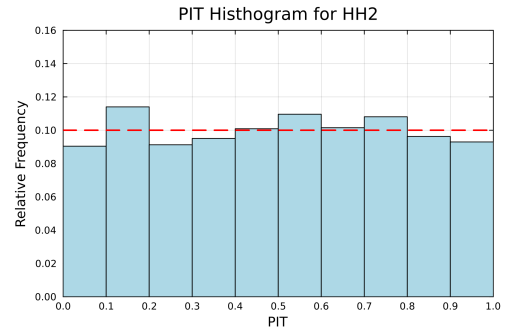


C. Results Model Analysis

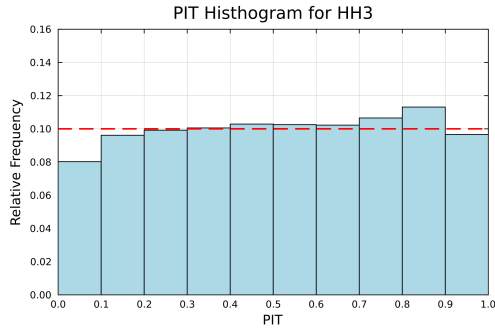
C.1. PIT Histograms



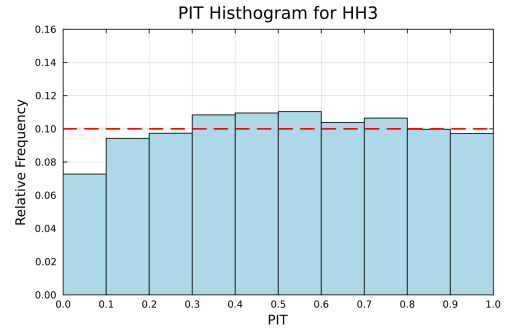
(a) Discretization Type A



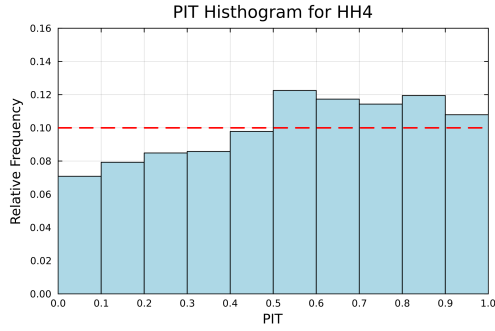
(b) Discretization Type B



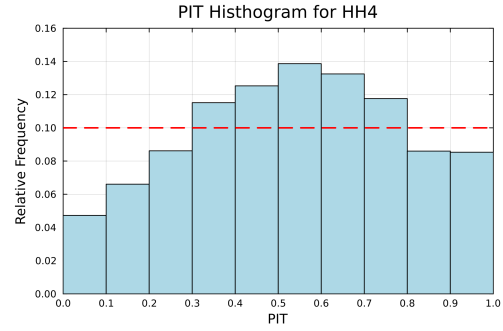
(a) Discretization Type A



(b) Discretization Type B

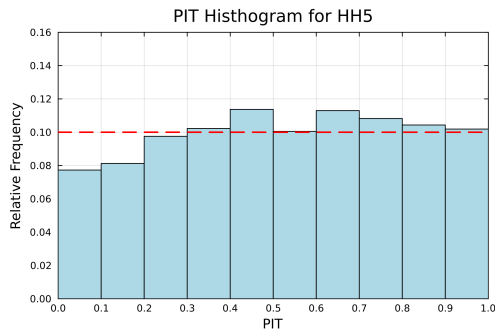


(a) Discretization Type A

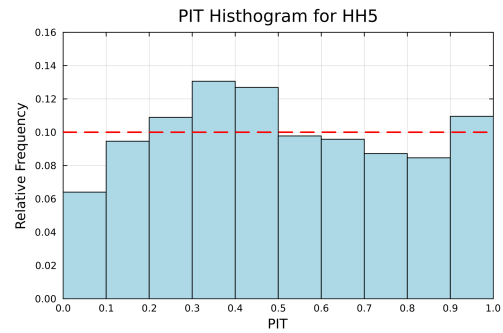


(b) Discretization Type B

Figure 13.: PIT Histograms for optimal models of HH4. While the model with equal-mass bins shows a slight skewness towards higher quantiles, the forecast of the model with equidistant bins is strongly overdispersed.

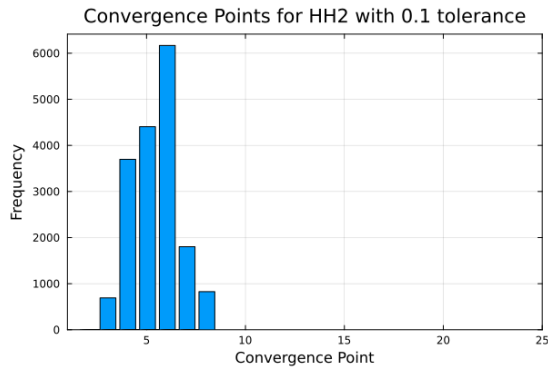


(a) Discretization Type A

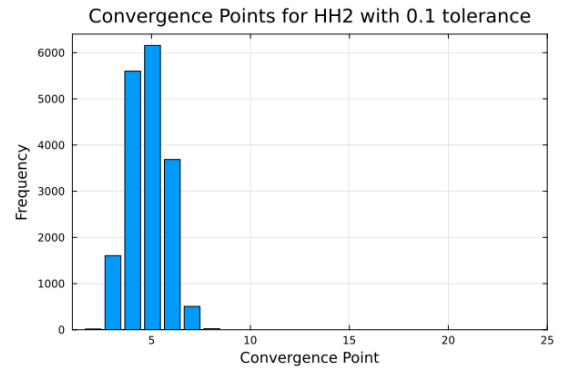


(b) Discretization Type B

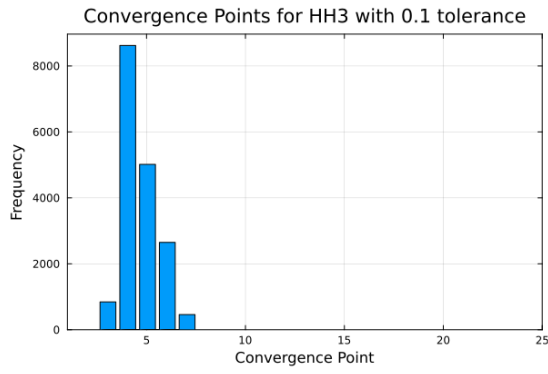
C.2. Tolerance 0.1



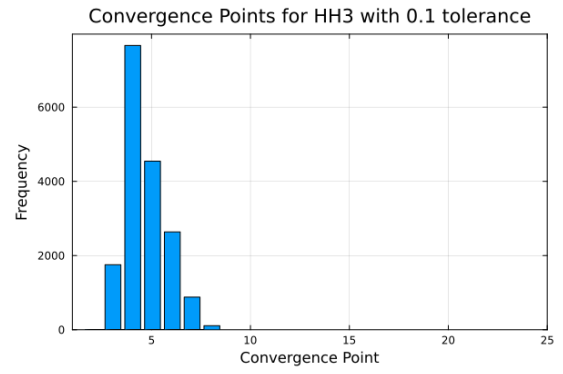
(a) Discretization Type A



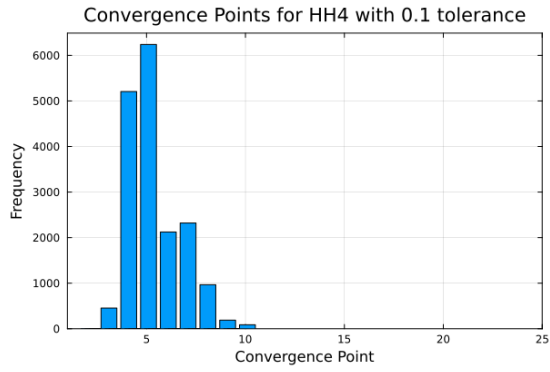
(b) Discretization Type B



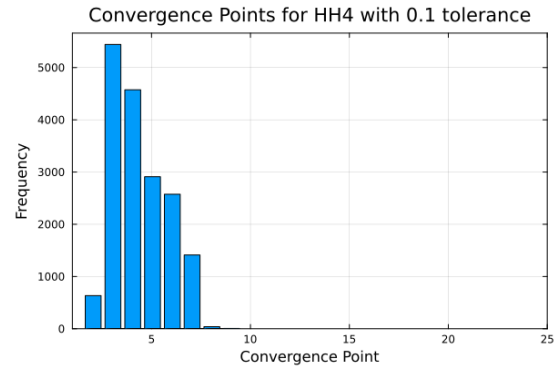
(a) Discretization Type A



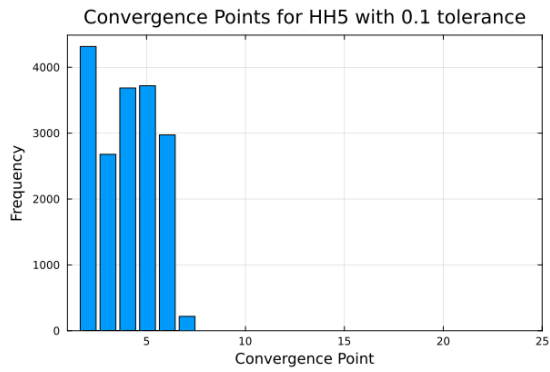
(b) Discretization Type B



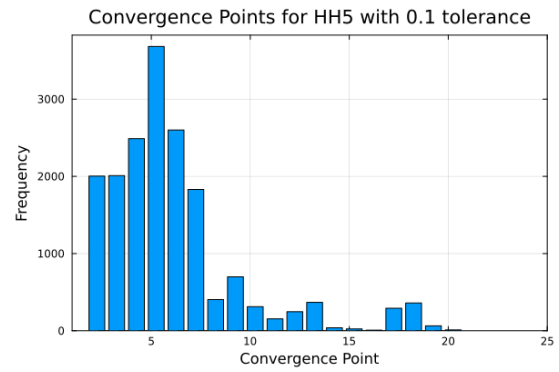
(a) Discretization Type A



(b) Discretization Type B

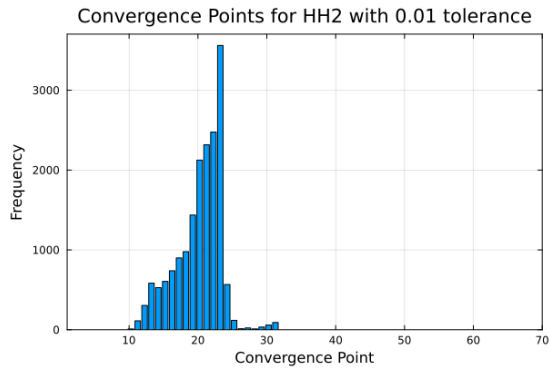


(a) Discretization Type A

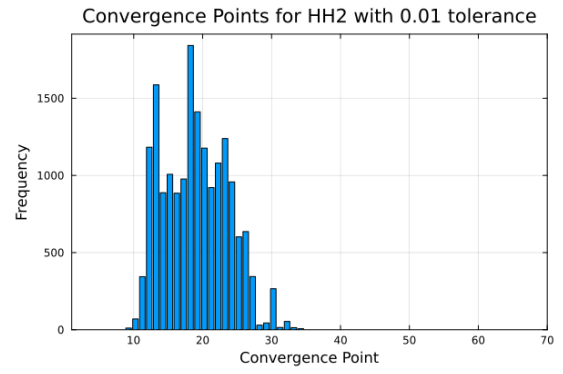


(b) Discretization Type B

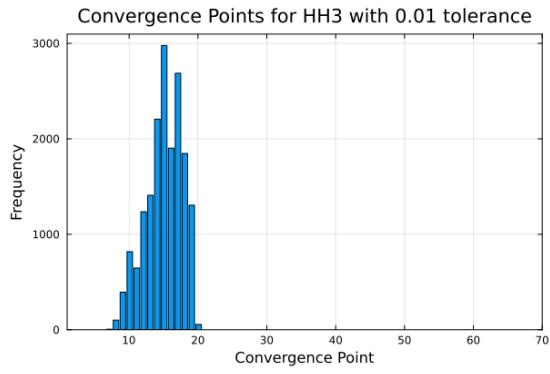
C.3. Tolerance 0.01



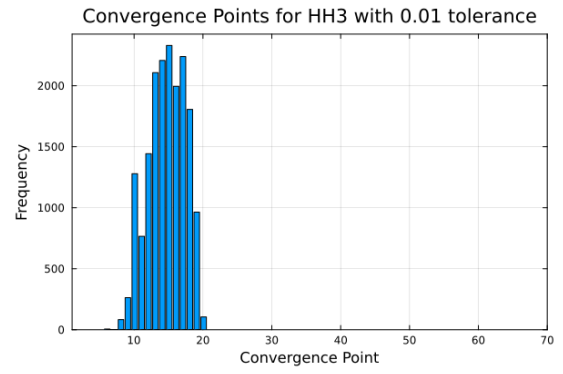
(a) Discretization Type A



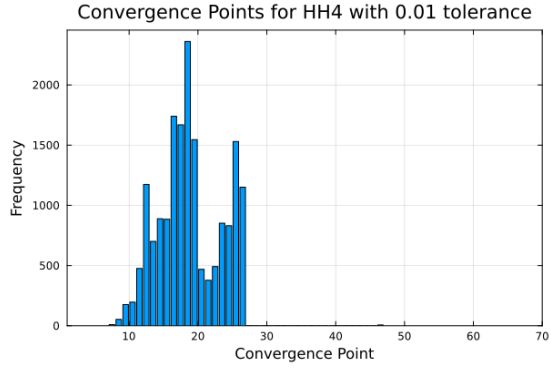
(b) Discretization Type B



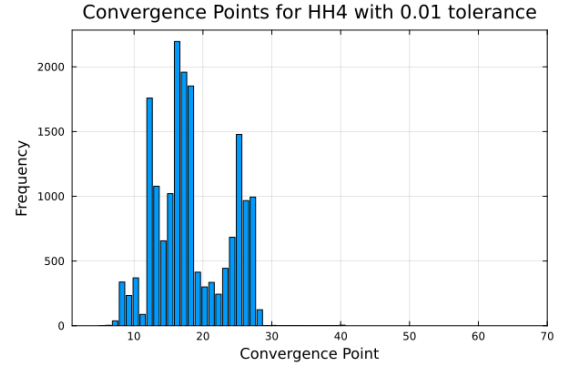
(a) Discretization Type A



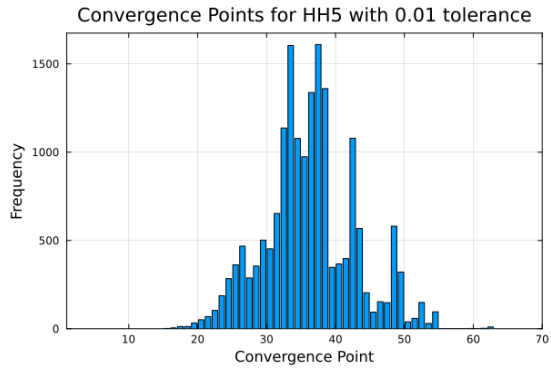
(b) Discretization Type B



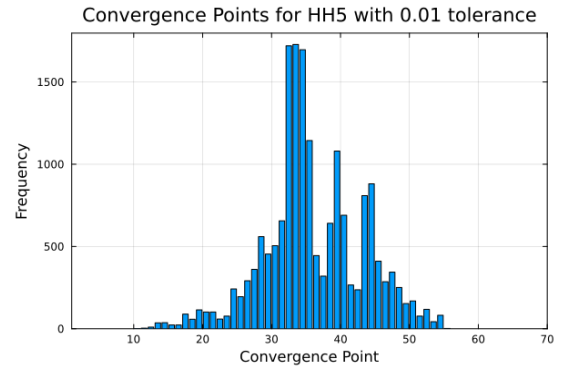
(a) Discretization Type A



(b) Discretization Type B



(a) Discretization Type A



(b) Discretization Type B